MTCWE - MikroTik Certified Wireless Engineer

Wireless Installations

Wireless routers
https://mikrotik.com/products/group/wireless-systems

Wireless cards



Antenna types

Types of WiFi antennas:

Omni-directional Antennas
These antennas send the signal in all directions around it. Think of these antennas as poles sending signals on a 360 degree circular plane. Omni-directional antennas are the default choice on most access points



Advantages:
- WiFi connection can be created in any direction
- Reduces effort in planning a wireless connection between remote buildings

Drawbacks:
- Since the signal is transmitted in all directions, it gets weaker with distance
- If all the receiving nodes are located in one direction, then the signal going in the other direction is wasted

Example of omni-directional antennas:
- Dipole antenna

Usage of omni-directional antennas:
- In enterprise
- In small office and home office (SOHO) WiFi deployment

Directional Antenna

A directional antenna sends out signal in a focussed way. These antennas direct coverage in a specific area in which they are mounted. They are the default option for outdoor PTP access point wireless network.



Advantages:

- Focuses the WiFi signal in a particular direction, thereby putting out high gain when compared to omni-directional antennas
- Decreases background noise or interference since the antenna will not hear the WiFi signal coming from any other direction except the ones coming from where it is pointing
- Increases the distance the signal will travel in one particular direction. This makes directional antenna a great choice for long distance links.

Drawback:

- To get the desired throughput, the link between the directional antenna needs to be planned well

Examples of directional antennas:

- Yagi antenna - beamwidth ranges between 30 - 80 degrees. Yagi antenna can provide >10 dBi passive gain
- Parabolic/Dish antenna - the beamwidth is usually <25 degrees; the antenna gain varies with diameter and frequency
- Grid antenna - a variation of the dish antenna

• Sector antenna - beamwidth ranges between 60 - 180 degrees, although 120 degrees is the typical beamwidth. In a sectorized array, these antennas can be mounted back-to-back for a complete 360 degree coverage

Wireless Standard

802.11 a/b/g/n

o Bands and channel width

o Frequencies

Protocol [edit]

IEEE 802.11 network PHY standards [hide]

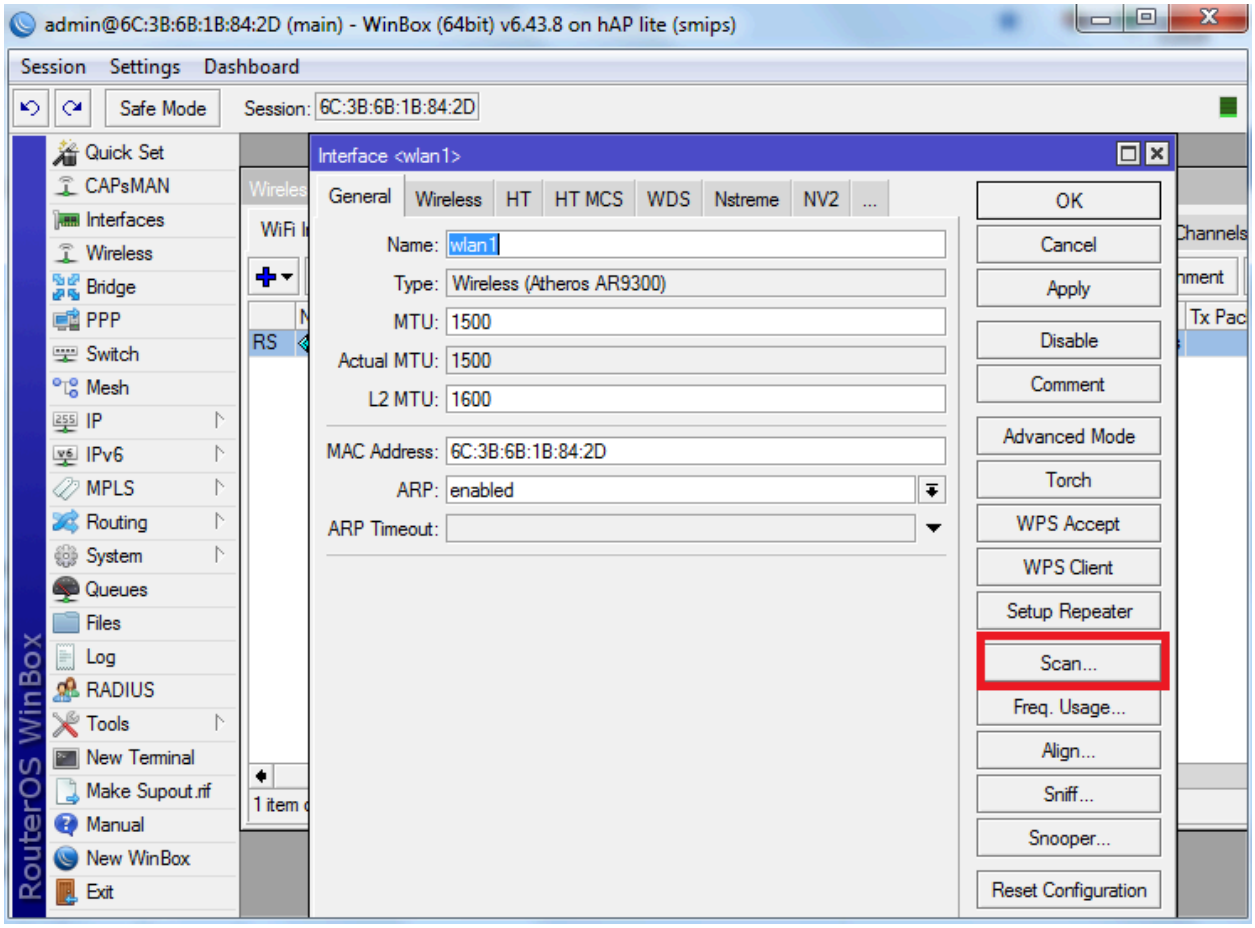
Frequency range, or type	PHY	Protocol	Release date[12]	Frequency (GHz)	Bandwidth (MHz)	Stream data rate[13] (Mbit/s)	Allowable MIMO streams	Modulation	Approximate range[citation needed] Indoor	Outdoor
1–6 GHz	DSSS/FHSS[14]	802.11-1997	Jun 1997	2.4	22	1, 2	N/A	DSSS, FHSS	20 m (66 ft)	100 m (330 ft)
	HR-DSSS[14]	802.11b	Sep 1999	2.4	22	1, 2, 5.5, 11	N/A	DSSS	35 m (115 ft)	140 m (460 ft)
	OFDM	802.11a	Sep 1999	5	5/10/20	6, 9, 12, 18, 24, 36, 48, 54 (for 20 MHz bandwidth, divide by 2 and 4 for 10 and 5 MHz)	N/A	OFDM	35 m (115 ft)	120 m (390 ft)
		802.11j	Nov 2004	4.9/5.0[D][15] [failed verification]					?	?
		802.11p	Jul 2010	5.9					?	1,000 m (3,300 ft)[16]
		802.11y	Nov 2008	3.7[A]					?	5,000 m (16,000 ft)[A]
	ERP-OFDM(, etc.)	802.11g	Jun 2003	2.4					38 m (125 ft)	140 m (460 ft)
	HT-OFDM[17]	802.11n	Oct 2009	2.4/5	20	Up to 288.8[B]	4	MIMO-OFDM	70 m (230 ft)	250 m (820 ft)[18] [failed verification]
					40	Up to 600[B]				
	VHT-OFDM[17]	802.11ac	Dec 2013	5	20	Up to 346.8[B]	8	MIMO-OFDM	35 m (115 ft)[19]	?
					40	Up to 800[B]				
					80	Up to 1733.2[B]				
					160	Up to 3466.8[B]				
	HE-OFDM	802.11ax	September 2019 [20]	2.4/5/6	20	Up to 1147[F]	8	MIMO-OFDM	30 m (98 ft)	120 m (390 ft)[G]
					40	Up to 2294[F]				
					80	Up to 4804[F]				
					80+80	Up to 9608[F]				

Wireless Tools

Scan

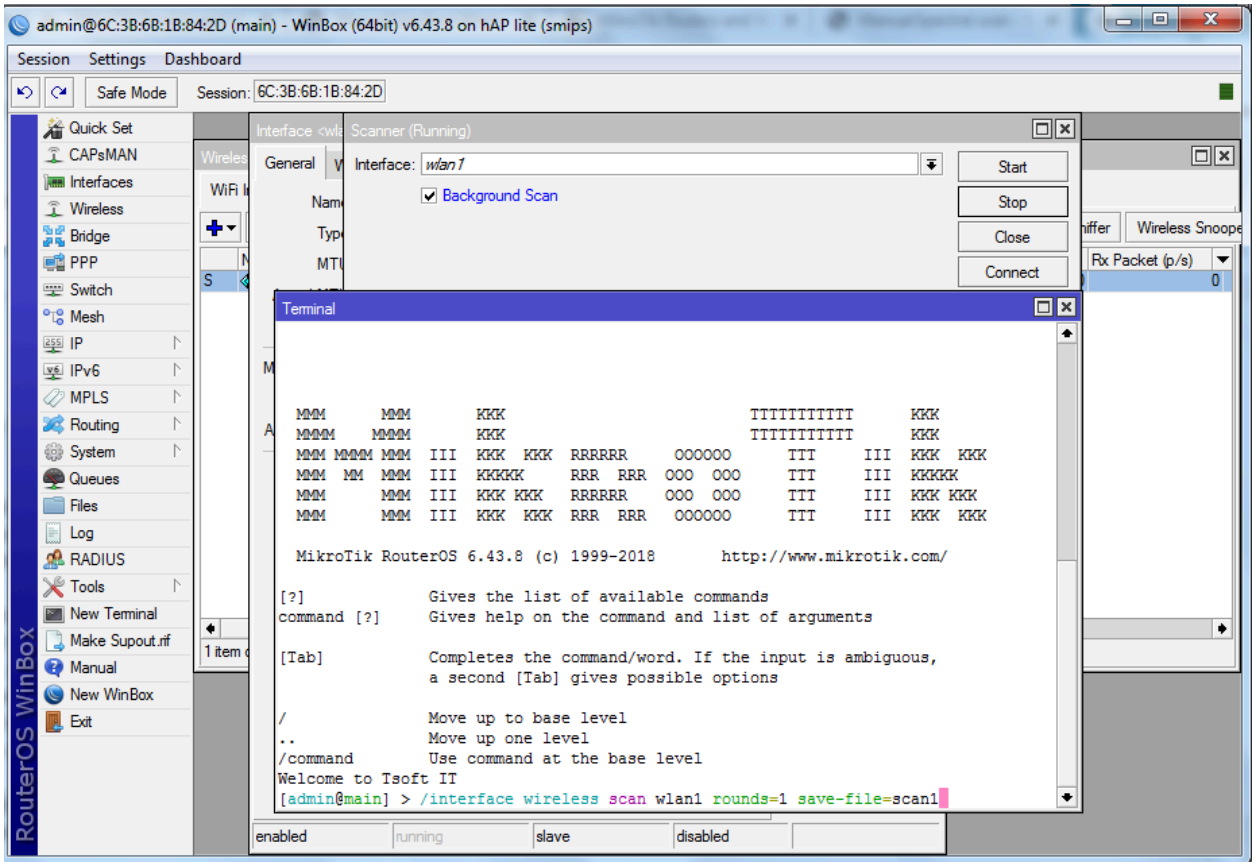
Scan tool will continue scanning for AP until user stops the scan process. It is possible to use 'rounds' setting for the scan tool to do scan through the scan-list entries specific times. It is useful when running scan tool using scripts. Example of scan command for one round:

/interface wireless scan wlan1 rounds=1



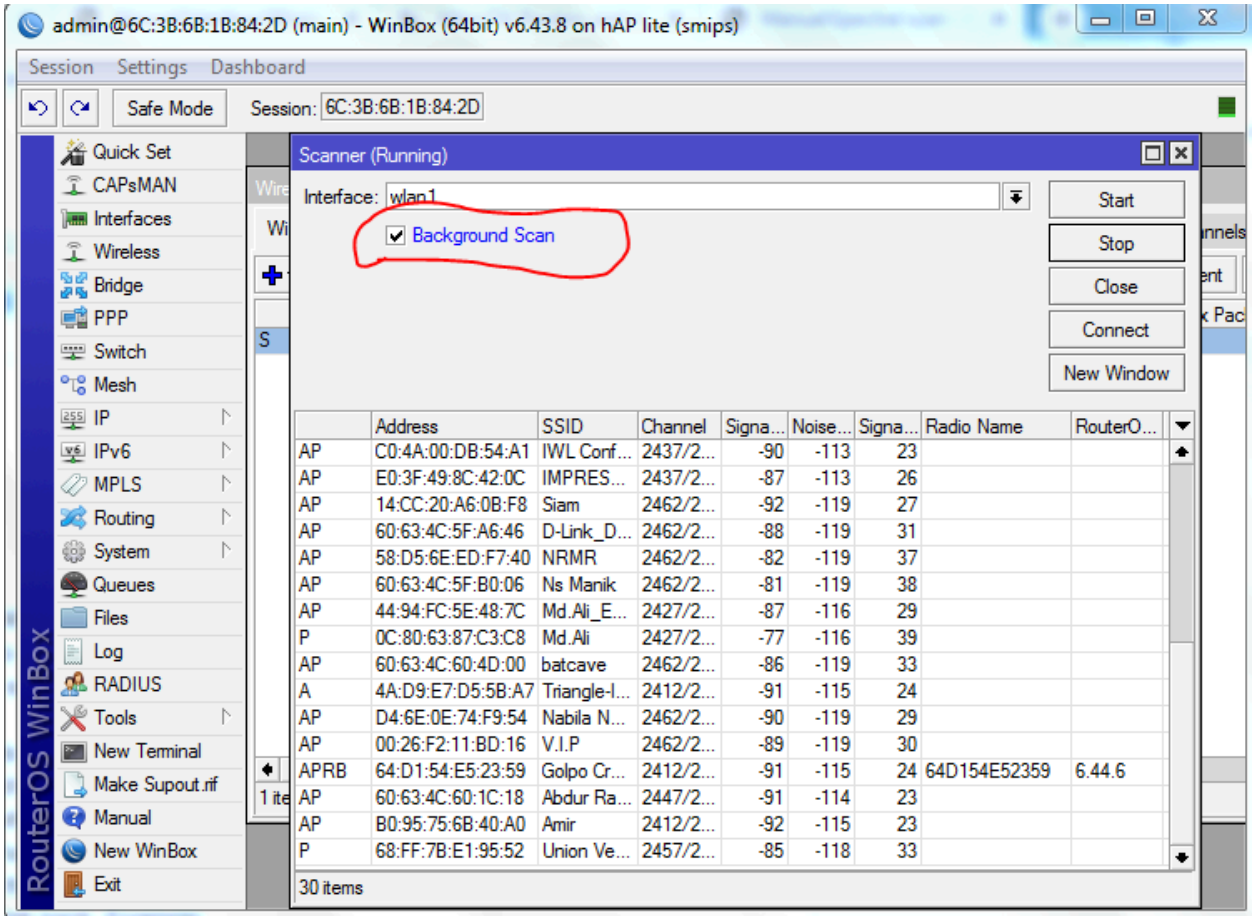
'save-file' option allows to do scripted/scheduled scans and save the results in file for future analysis. Also this feature together with rounds setting allows to get scan results from the remote wireless clients - executing that command will start the scan tool which disconnect the wireless link, does the scan through the scan-list frequencies and saves the results to file, exits the scan and connects the wireless link back. Example:

/interface wireless scan wlan1 rounds=1 save-file=scan1



To use background wireless scan the 'background=yes' setting should be provided. Example:

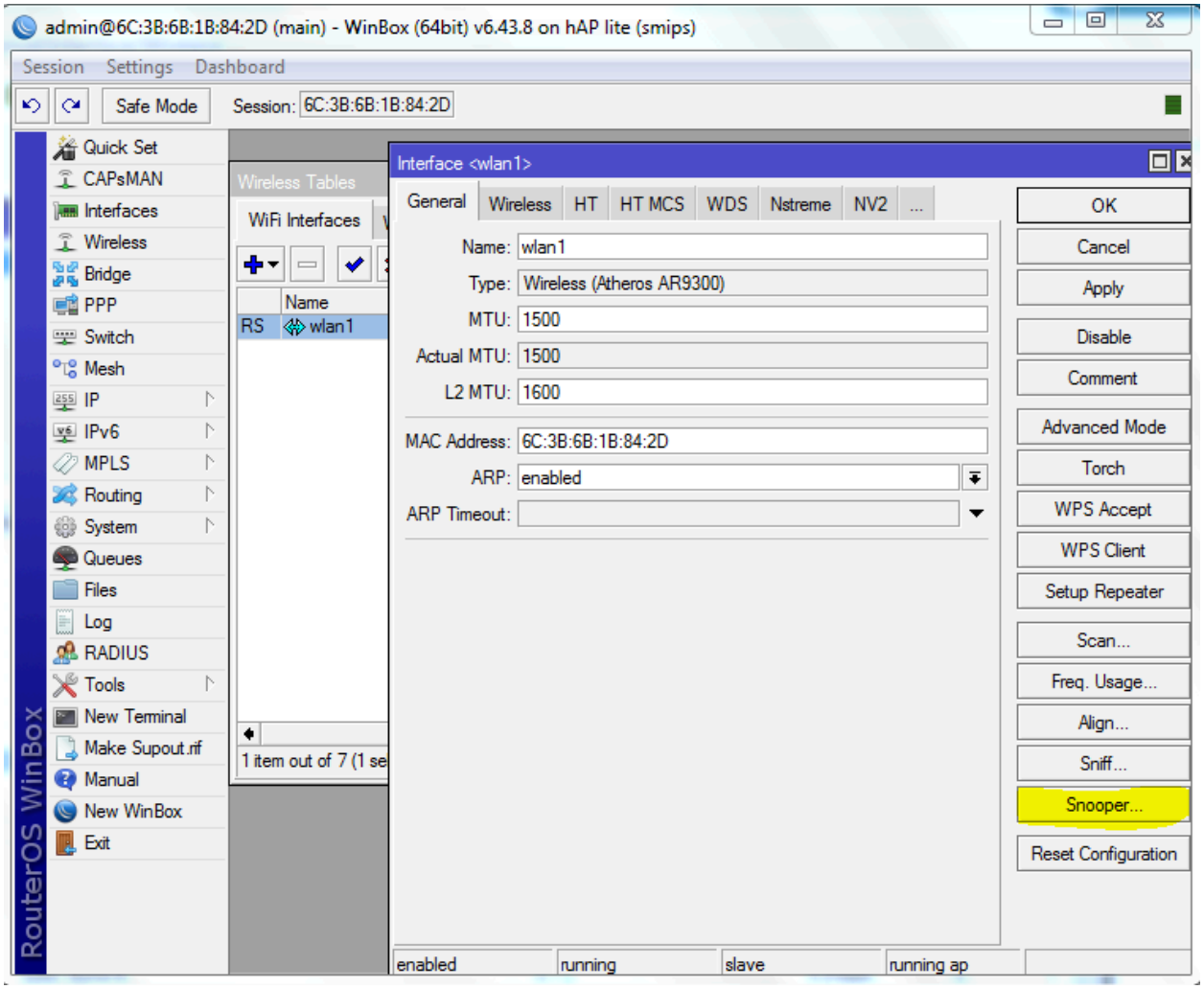
/interface wireless scan wlan1 background=yes



Snooper/Frequency usage

This tool monitors surrounding frequency usage, and displays which devices occupy each frequency. It's available both in console, and also in Winbox.

Sub-menu: /interface wireless snooper

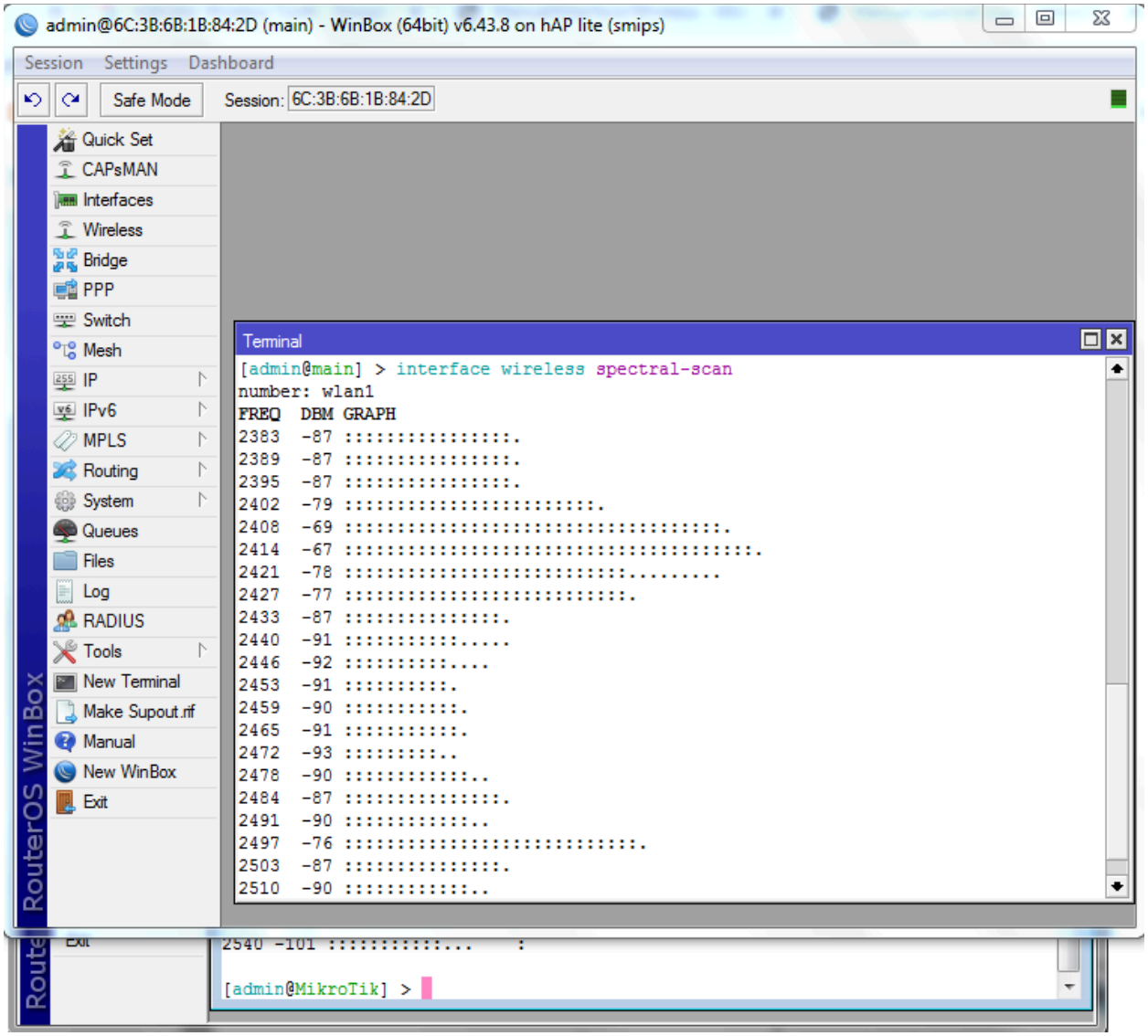


Spectral scan

The spectral scan can scan all frequencies supported by your wireless card, and plot them directly in console. Exact frequency span depends on card. Allowed ranges on r52n: [4790; 6085], [2182; 2549].

Wireless card can generate 4us long spectral snapshots for any 20mhz wide channel. This is considered a single spectral sample.

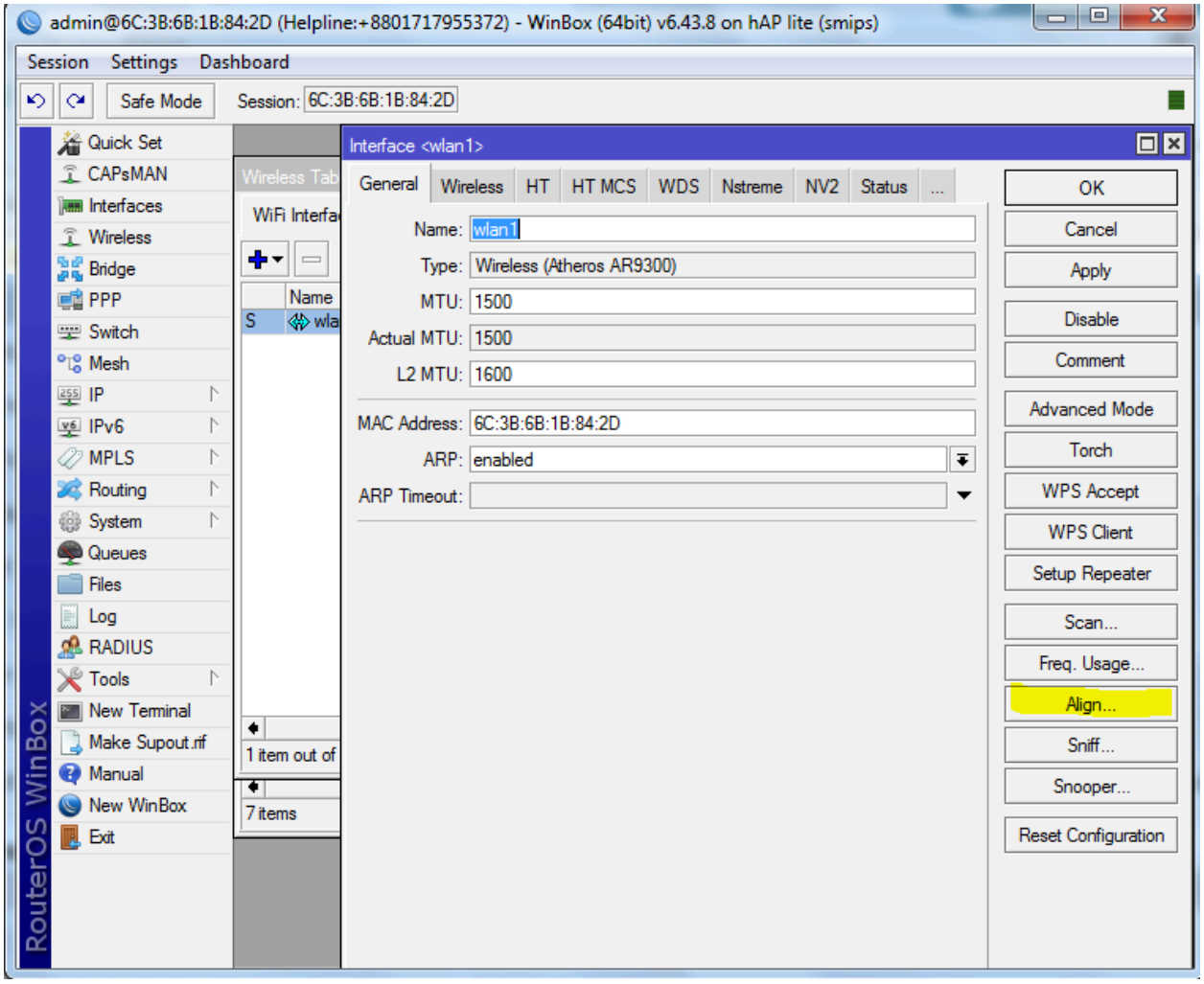
To improve data quality spectrum is scanned with 10mhz frequency increments, which means doubled sample coverage at each specific frequency (considering 20mhz wide samples).



Align

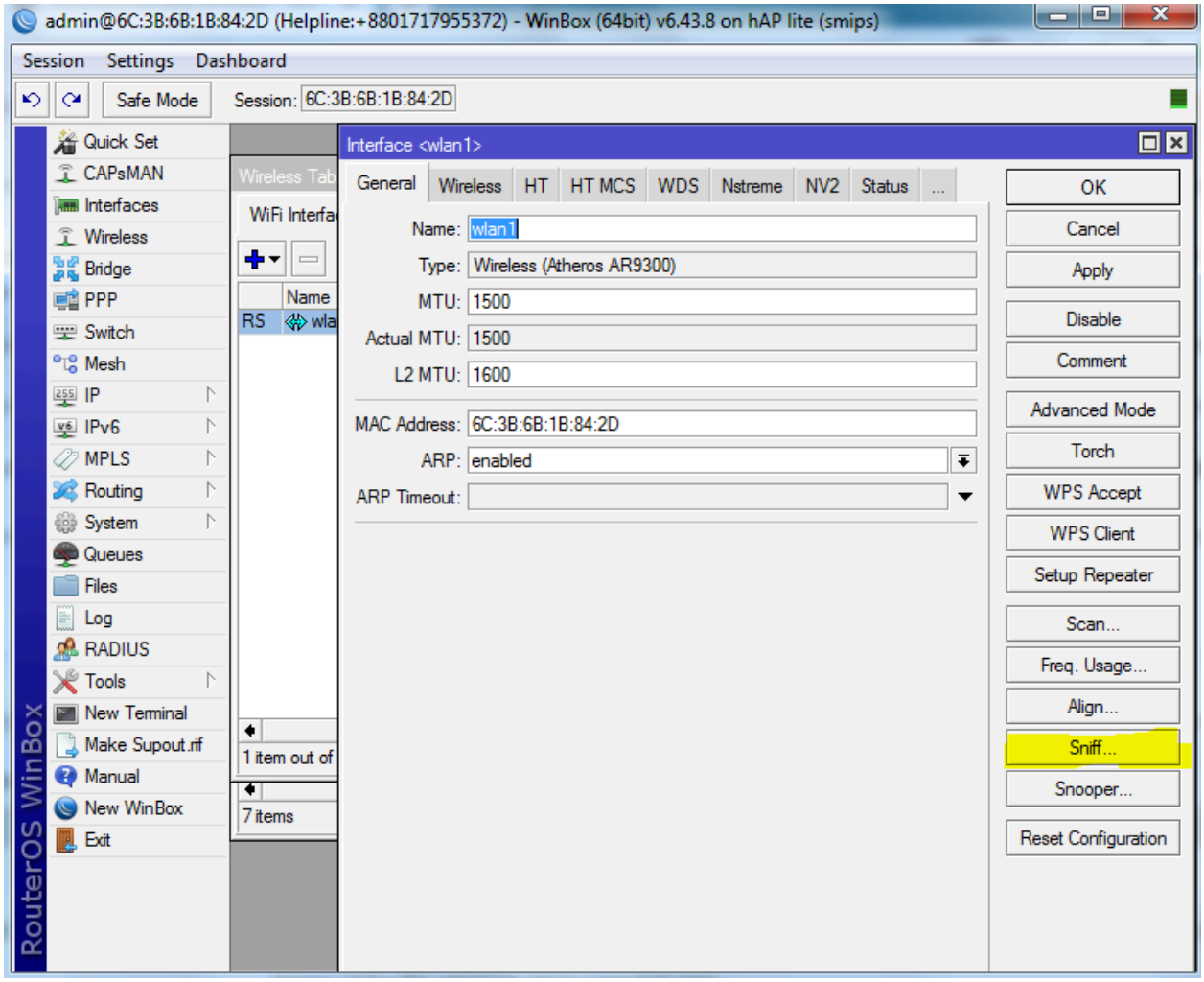
/interface wireless align

Align tool is used to help in alignment devices running this tool.



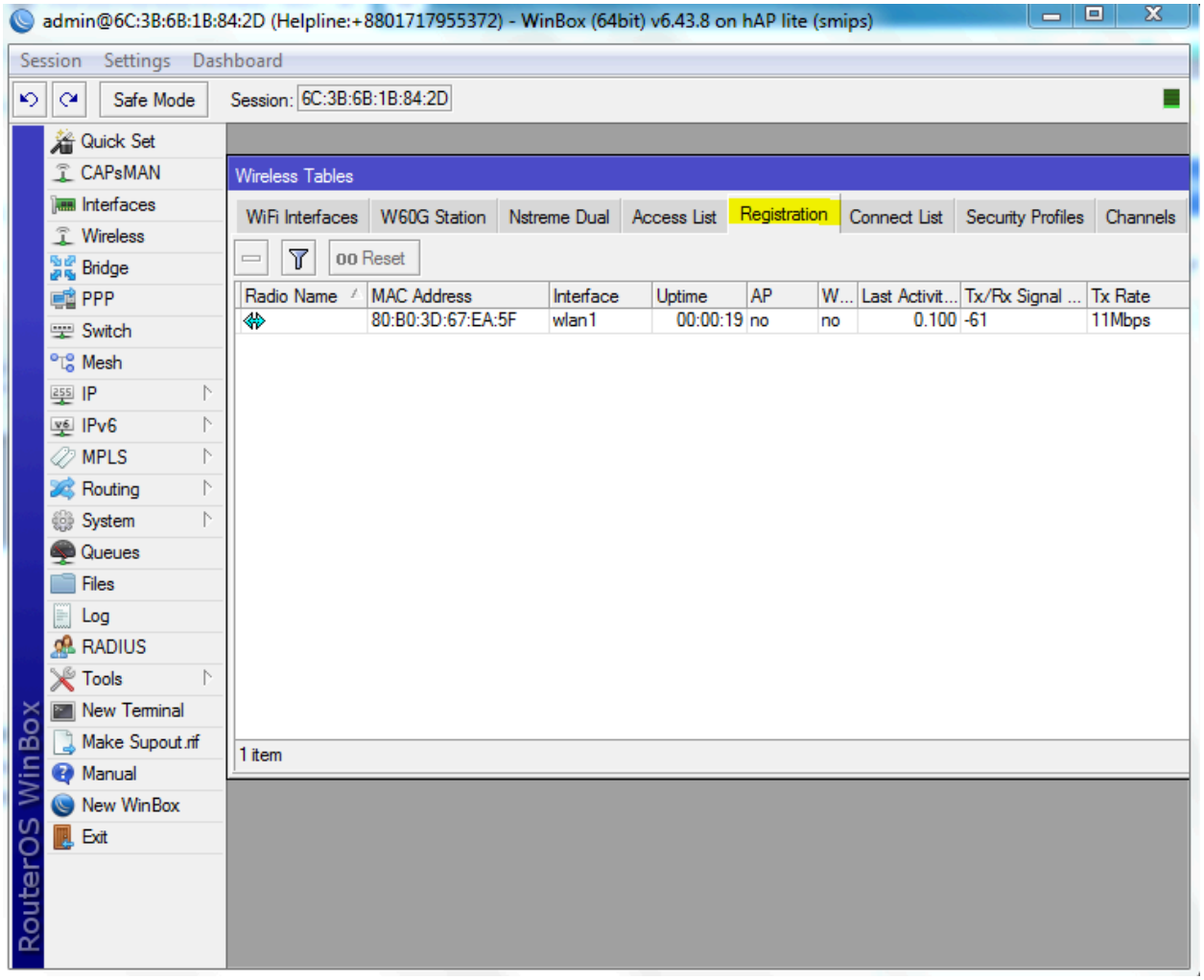
sniffer
/interface wireless sniffer

Wireless sniffer allows to capture frames including Radio header, 802.11 header and other wireless related information.



Wireless Troubleshooting

Registration Table
/interface wireless registration-table
In the registration table, you can see various information about currently connected clients. It is used only for Access Points.



Ack-Timeout/Distance

How long to wait for confirmation of unicast frames (ACKs) before considering transmission unsuccessful, or in short ACK-Timeout. Distance value has these behaviors:

Dynamic - causes AP to detect and use the smallest timeout that works with all connected clients.

Indoor - uses the default ACK timeout value that the hardware chip manufacturer has set.

Number - uses the input value in formula: ACK-timeout = ((distance * 1000) + 299) / 300 us;

Acknowledgments are not used in Nstreme/NV2 protocols.

CCQ

CCQ stands for Client Connection Quality. CCQ two types

tx-ccq

rx-ccq

Client Connection Quality (CCQ) is a value in percent that shows how effective the bandwidth is used regarding the theoretically maximum available bandwidth. CCQ is weighted average of

values Tmin/Treal, that get calculated for every transmitted frame, where Tmin is time it would take to transmit given frame at highest rate with no retries and Treal is time it took to transmit frame in real life (taking into account necessary retries it took to transmit frame and transmit rate).

TX/RX Signal Strength

Tx and Rx Levels. The Tx level is the power in decibels per milliwatt (dBm) at which a modem transmits its signal.
The Rx level is the power in dBm of the received signal. The server modems normally transmit at -13 dBm (decibels per milliwatt) by default.
Ideally, the Rx level should be in the range of -18 to -25 dBm.

Frames and HW-frames

The frames counts only those which contain actual data.
In case of Nstreme, only the ACK can be transmitted in a single frame, if there is no other data to send. These ACK frames will not be added to the frames count, but they will appear at hw-frames. If there is traffic on both directions at maximum speed (eg. there will be no only-ack frames), then you can't compare frames to hw-frames as in case of regular wireless links.

Data-rates

IEEE 802.11 Wi-Fi protocol summary

Protocol	Frequency	Channel Width	MIMO	Maximum data rate (theoretical)
802.11ax	2.4 or 5GHz	20, 40, 80, 160MHz	Multi User (MU-MIMO)	2.4 Gbps[1]
802.11ac wave2	5 GHz	20, 40, 80, 160MHz	Multi User (MU-MIMO)	1.73 Gbps[2]
802.11ac wave1	5 GHz	20, 40, 80MHz	Single User (SU-MIMO)	866.7 Mbps[2]
802.11n	2.4 or 5 GHz	20, 40MHz	Single User (SU-MIMO)	450 Mbps[3]
802.11g	2.4 GHz	20 MHz	N/A	54 Mbps
802.11a	5 GHz	20 MHz	N/A	54 Mbps
802.11b	2.4 GHz	20 MHz	N/A	11 Mbps
Legacy 802.11	2.4 GHz	20 MHz	N/A	2 Mbps

Wireless Advanced Settings

What is hw-retries setting?

Number of times sending frame is retried without considering it a transmission failure. The data rate is decreased upon failure and frame is sent again. Three sequential failures on lowest supported rate suspend transmission to this destination for the duration of on-fail-retry-time. After that, the frame is sent again. The frame is being retransmitted until transmission success, or until the client is disconnected after disconnect-timeout. The frame can be discarded during

this time if frame-lifetime is exceeded. In case of Nstreme "on-fail-retry-time", "disconnect-timeout" and "frame-lifetime" settings are not used. So after three sequential failures on the lowest supported rate, the wireless link is disconnected with "extensive data loss" message.

HW-protection

Frame protection support (RTS/CTS)

802.11 standard provides means to protect transmission against other device transmission by using RTS/CTS protocol. Frame protection helps to fight "hidden node" problem. There are several types of protection:

RTS/CTS based protection - device willing to send frame at first sends RequestToSend frame and waits for ClearToSend frame from intended destination. By "seeing" RTS or CTS frame 802.11 compliant devices know that somebody is about to transmit and therefore do not initiate transmission themselves

"CTS to self" based protection - device willing to send frame sends CTS frame "to itself". As in RTS/CTS protocol every 802.11 compliant device receiving this frame know not to transmit. "CTS to self" based protection has less overhead, but it must be taken into account that this only protects against devices receiving CTS frame (e.g. if there are 2 "hidden" stations, there is no use for them to use "CTS to self" protection, because they will not be able to receive CTS sent by other station - in this case stations must use RTS/CTS so that other station knows not to transmit by seeing CTS transmitted by AP).

Protection mode is controlled by hw-protection-mode setting of wireless interface. Possible values: none - for no protection (default), rts-cts for RTS/CTS based protection or cts-to-self for "CTS to self" based protection.

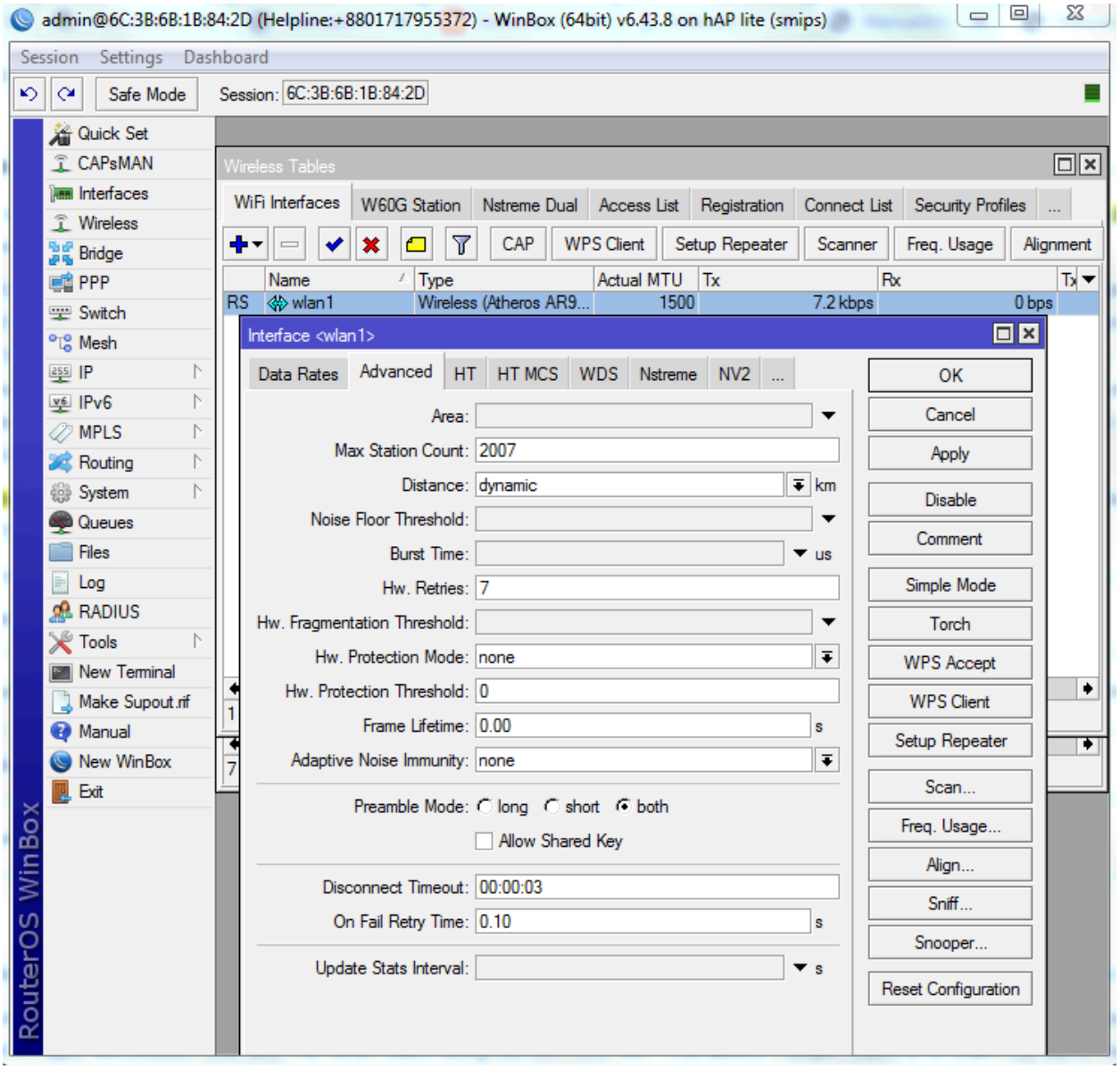
Frame size threshold at which protection should be used is controlled by hw-protection-threshold setting of wireless interface.

For example, to enable "CTS-to-self" based frame protection on AP for all frames, not depending on size, use command:

```
[admin@MikroTik] /interface wireless> set 0 hw-protection-mode=cts-to-self
hw-protection-threshold=0
```

To enable RTS/CTS based protection on client use command:

```
[admin@MikroTik] /interface wireless> set 0 hw-protection-mode=rts-cts
hw-protection-threshold=0
```



What is adaptive-noise-immunity setting?
Adaptive Noise Immunity (ANI) adjusts various receiver parameters dynamically to minimize interference and noise effect on the signal quality [1] This setting is added in the wireless driver for Atheros AR5212 and newer chipset cards

WMM
What is WMM (WiFi Multimedia)?

WMM maintains the priority of audio, video and voice, over other applications which are less time critical.Using QoS (Quality of Service) Wi-Fi Multi-Media (WMM) ensures that the applications that require better throughput and performance,are inserted in queues with higher

priority. For example, video and audio applications are given higher priority over applications such as FTP.

In this way, in a phone conversation, you are less likely to hear delays. Watching video, you are more likely to see smooth action. This delays other network traffic of a less critical nature - such as downloading large files - where a small time delay is acceptable.

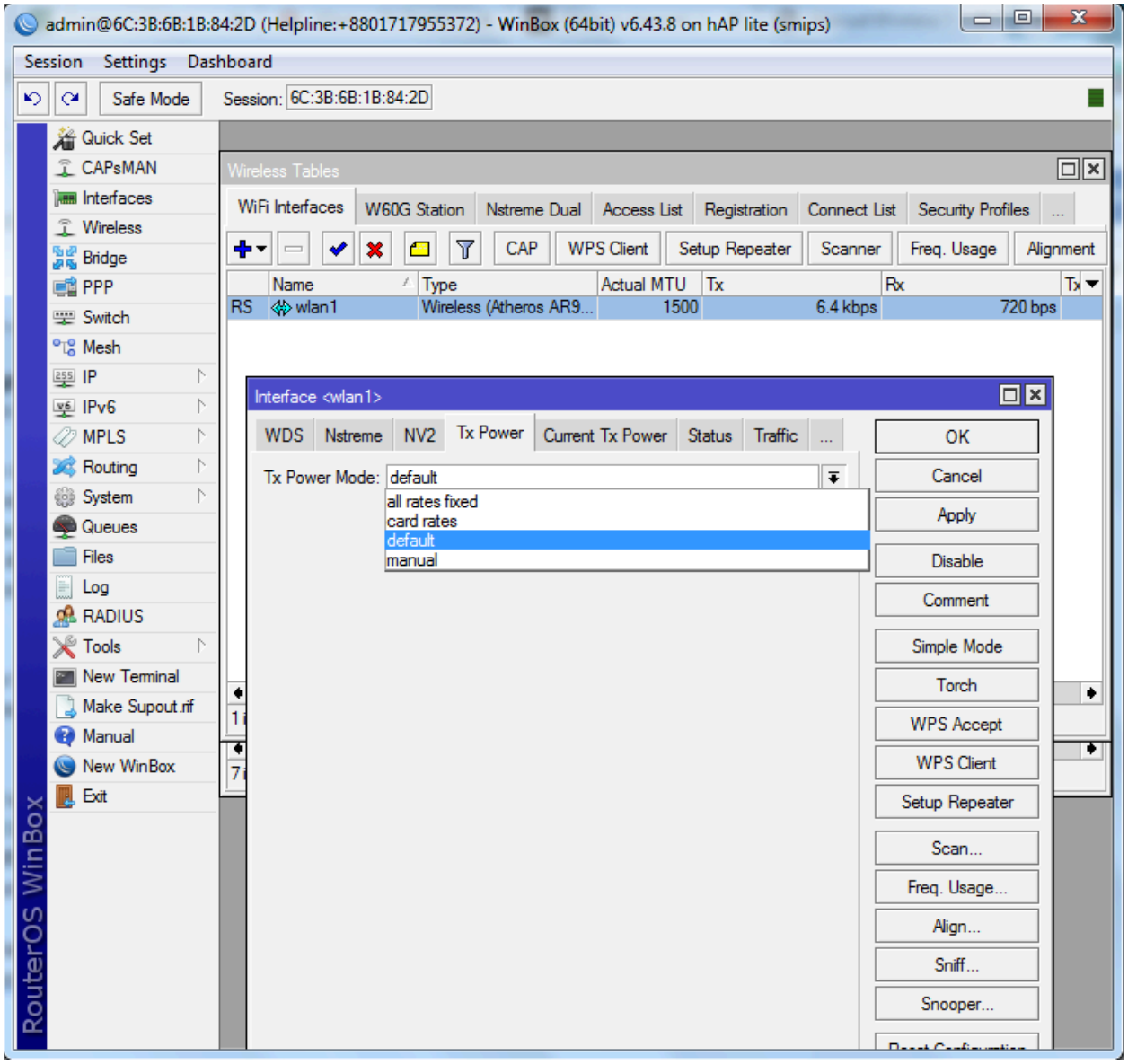
WMM defines the following four queues, in decreasing order of priority:

Voice.The highest priority queue with minimum delay, which makes it ideal for applications such as voice over IP(VoIP) and streaming media.
Video. The second highest priority queue with low delay is given to this queue. Video applications are routed to this queue.
Best Effort. The medium priority queue with medium delay is given to this queue. Most standard IP applications use this queue.
Background. Low priority queue with high throughput. Applications, such as FTP, that are not time-sensitive, but require high throughput,can use this queue.

Country regulation settings
Antenna gain in dBi, used to calculate maximum transmit power according to country regulations.
Three frequency modes are available:
regulatory-domain - Limit available channels and maximum transmit power for each channel according to the value of country
manual-txpower - Same as above, but do not limit maximum transmit power.
superchannel - Conformance Testing Mode. Allow all channels supported by the card

TX-power
What TX-power-mode is the best?
TX-power-mode tells the wireless card which tx-power values should be used. By default, this setting is set to default.

default means that the card will use the tx-power values from the cards eeprom and will ignore the setting what is specified by the user in the tx-power field.
card-rates means that for different data rates the tx-power is calculated according to the cards transmit power algorithm from the cards eeprom and as an argument it takes tx-power value specified by the user.
all-rates-fixed means that that the card will use one tx-power value for all data rates which is specified by the user in tx-power field.



Virtual-AP

Check virtual AP lecture sheet

802.11n

802.11n is an amendment that improves upon the previous 802.11 standards, that had the first draft of certification published in 2006. The 802.11n standard was retroactively labelled as Wi-Fi 4 by the Wi-Fi Alliance. The standard added support for multiple-input multiple-output antennas (MIMO). 802.11n operates on both the 2.4 GHz and the 5 GHz bands. Support for 5 GHz bands is optional. Its net data rate ranges from 54 Mbit/s to 600 Mbit/s. The IEEE has approved the amendment, and it was published in October 2009. Prior to the final ratification, enterprises

were already migrating to 802.11n networks based on the Wi-Fi Alliance's certification of products conforming to a 2007 draft of the 802.11n proposal.

Number of antennas

The number of simultaneous data streams is limited by the minimum number of antennas in use on both sides of the link. However, the individual radios often further limit the number of spatial streams that may carry unique data. The a x b : c notation helps identify what a given radio is capable of. The first number (a) is the maximum number of transmit antennas or TX RF chains that can be used by the radio. The second number (b) is the maximum number of receive antennas or RX RF chains that can be used by the radio. The third number (c) is the maximum number of data spatial streams the radio can use. For example, a radio that can transmit on two antennas and receive on three, but can only send or receive two data streams would be 2 x 3 : 2.

The 802.11n draft allows up to 4 x 4 : 4. Common configurations of 11n devices are 2x2:2, 2x3:2, and 3x2:2. All three configurations have the same maximum throughputs and features, and differ only in the amount of diversity the antenna systems provide. In addition, a fourth configuration, 3x3:3 is becoming common, which has a higher throughput, due to the additional data stream.[9]

Data rates

Assuming equal operating parameters to an 802.11g network achieving 54 megabits per second (on a single 20 MHz channel with one antenna), an 802.11n network can achieve 72 megabits per second (on a single 20 MHz channel with one antenna and 400 ns guard interval); 802.11n's speed may go up to 150 megabits per second if there are not other Bluetooth, microwave or Wi-Fi emissions in the neighborhood by using two 20 MHz channels in 40 MHz mode. If more antennas are used, then 802.11n can go up to 288 megabits per second in 20 MHz mode with four antennas, or 600 megabits per second in 40 MHz mode with four antennas and 400 ns guard interval. Because the 2.4 GHz band is seriously congested in most urban areas, 802.11n networks usually have more success in increasing data rate by utilizing more antennas in 20 MHz mode rather than by operating in the 40 MHz mode, as the 40 MHz mode requires a relatively free radio spectrum which is only available in rural areas away from cities. Thus, network engineers installing an 802.11n network should strive to select routers and wireless clients with the most antennas possible (one, two, three or four as specified by the 802.11n standard) and try to make sure that the network's bandwidth will be satisfactory even on the 20 MHz mode.

Data rates up to 600 Mbit/s are achieved only with the maximum of four spatial streams using one 40 MHz-wide channel. Various modulation schemes and coding rates are defined by the standard and are represented by a Modulation and Coding Scheme (MCS) index value. The table below shows the relationships between the variables that allow for the maximum data rate. GI (Guard Interval): Timing between symbols.[10]

20 MHz channel uses an FFT of 64, of which: 56 OFDM subcarriers, 52 are for data and 4 are pilot tones with a carrier separation of 0.3125 MHz (20 MHz/64) (3.2 μs). Each of these subcarriers can be a BPSK, QPSK, 16-QAM or 64-QAM. The total bandwidth is 20 MHz with an occupied bandwidth of 17.8 MHz. Total symbol duration is 3.6 or 4 microseconds, which includes a guard interval of 0.4 or 0.8 microseconds.

Frame aggregation
PHY level data rate does not match user level throughput because of 802.11 protocol overheads, like the contention process, interframe spacing, PHY level headers (Preamble + PLCP) and acknowledgment frames. The main media access control (MAC) feature that provides a performance improvement is aggregation. Two types of aggregation are defined:

Aggregation of MAC service data units (MSDUs) at the top of the MAC (referred to as MSDU aggregation or A-MSDU)
Aggregation of MAC protocol data units (MPDUs) at the bottom of the MAC (referred to as MPDU aggregation or A-MPDU)
Frame aggregation is a process of packing multiple MSDUs or MPDUs together to reduce the overheads and average them over multiple frames, thereby increasing the user level data rate. A-MPDU aggregation requires the use of block acknowledgement or BlockAck, which was introduced in 802.11e and has been optimized in 802.11n.

Backward compatibility
When 802.11g was released to share the band with existing 802.11b devices, it provided ways of ensuring coexistence between legacy and successor devices. 802.11n extends the coexistence management to protect its transmissions from legacy devices, which include 802.11g, 802.11b and 802.11a. There are MAC and PHY level protection mechanisms as listed below:

PHY level protection: Mixed Mode Format protection (also known as L-SIG TXOP Protection): In mixed mode, each 802.11n transmission is always embedded in an 802.11a or 802.11g transmission. For 20 MHz transmissions, this embedding takes care of the protection with 802.11a and 802.11g. However, 802.11b devices still need CTS protection.
PHY level protection: Transmissions using a 40 MHz channel in the presence of 802.11a or 802.11g clients require using CTS protection on both 20 MHz halves of the 40 MHz channel, to prevent interference with legacy devices.
MAC level protection: An RTS/CTS frame exchange or CTS frame transmission at legacy rates can be used to protect subsequent 11n transmission.

Setup and Testing
This section details how to configure your wireless 802.11n card along with an example at the end of a preconfigured setup you can attempt on your own cards. I will also go into as much

detail as possible on the new options provided to the 802.11n cards and what each of these does.

802.11n Configuration options
Winbox changable configuration options
band (2ghz-b/g/n | 2ghz-onlyn | 5ghz-a/n | 5ghz-onlyn | other-original a/b/g options also available)
ht-extension-channel (above-control | below-control | disabled)
ht-rxchains (0,1,2 - any combination of these)
ht-txchains (0,1,2 - any combination of these)
ht-ampdu-priorities (0,1,2,3,4,5,6,7 - any combination of these)
ht-guard-interval (any | long) [NV2 only supports 800nS long guard interval!]
These items can be viewed and changed from in winbox or terminal.

Winbox viewable configuration options
ht-basic-mcs
(mcs-0,mcs-1,mcs-2,mcs-3,mcs-4,mcs-5,mcs-6,mcs-7,mcs-8,mcs-9,mcs-10,mcs-11,mcs-12,mcs-13,mcs-14,mcs-15 - any combination of these)
ht-supported-mcs
(mcs-0,mcs-1,mcs-2,mcs-3,mcs-4,mcs-5,mcs-6,mcs-7,mcs-8,mcs-9,mcs-10,mcs-11,mcs-12,mcs-13,mcs-14,mcs-15 - any combination of these)
These previous 2 options can be changed only from terminal, but appear in winbox. (In ROS4.0beta4 they can be changed in winbox Wireless Interface window, tab "Data Rates" and set rate to "configured". Under "HT MCS" tab options are now configurable.)

Non-Winbox viewable configuration options
ht-amsdu-limit (0..8192)
ht-amsdu-threshold (0..8192)
These last two only appear in terminal at this time. (ROS4.0beta4 made these configurable in winbox too.)

Wireless Security
Access Management
Wireless security is the prevention of unauthorized access or damage to computers or data using wireless networks, which include Wi-Fi networks. The most common type is Wi-Fi security, which includes Wired Equivalent Privacy (WEP) and Wi-Fi Protected Access (WPA). WEP is a notoriously weak security standard[citation needed]: the password it uses can often be cracked in a few minutes with a basic laptop computer and widely available software tools. WEP is an old IEEE 802.11 standard from 1997,[1] which was superseded in 2003 by WPA, or Wi-Fi Protected Access. WPA was a quick alternative to improve security over WEP. The current standard is WPA2; some hardware cannot support WPA2 without firmware upgrade or replacement. WPA2 uses an encryption device that encrypts the network with a 256-bit key; the

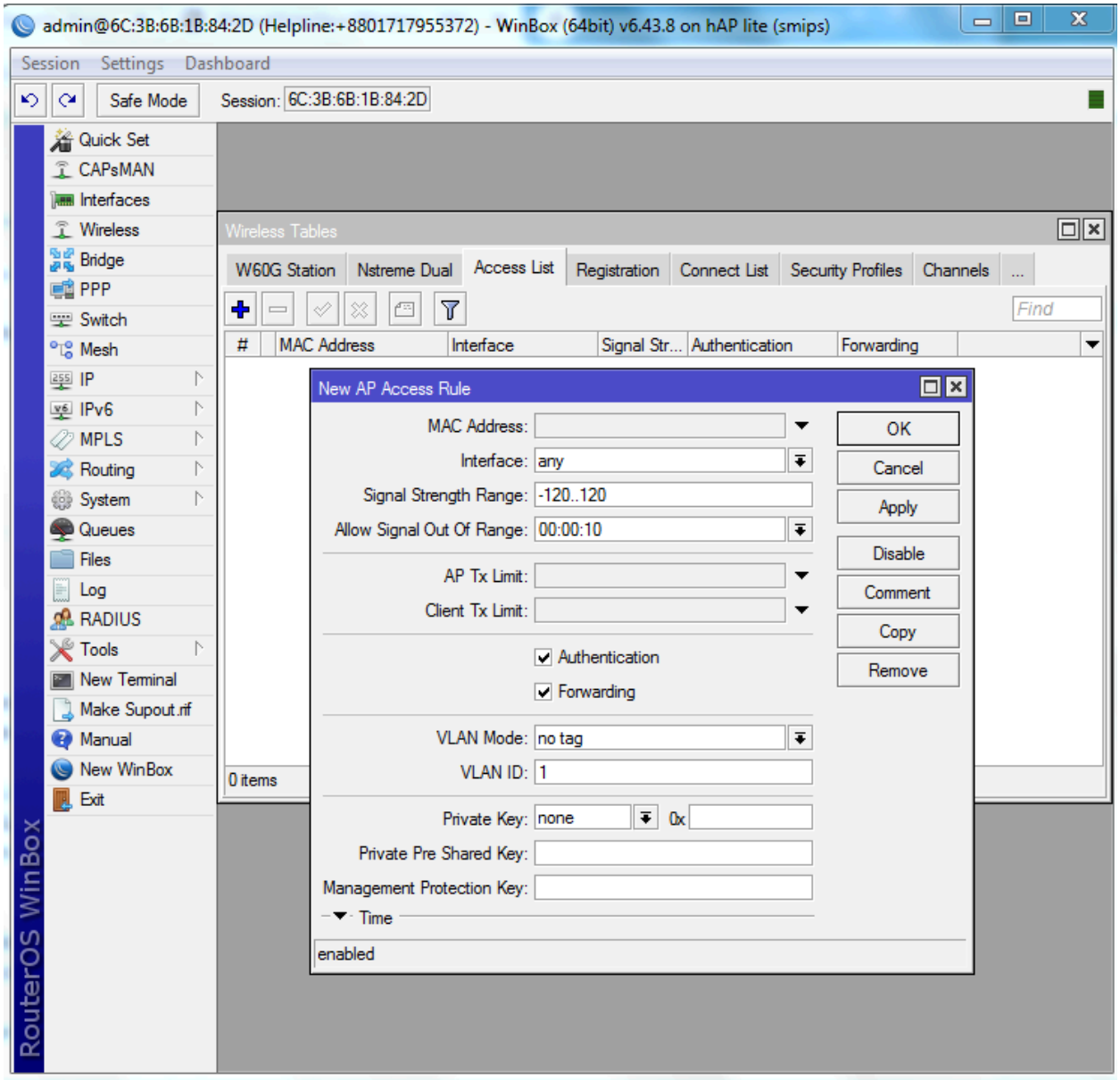
longer key length improves security over WEP. Enterprises often enforce security using a certificate-based system to authenticate the connecting device, following the standard 802.1X.

Access-List/Connect-List
Access List
Sub-menu: /interface wireless access-list

Access list is used by access point to restrict allowed connections from other devices, and to control connection parameters.

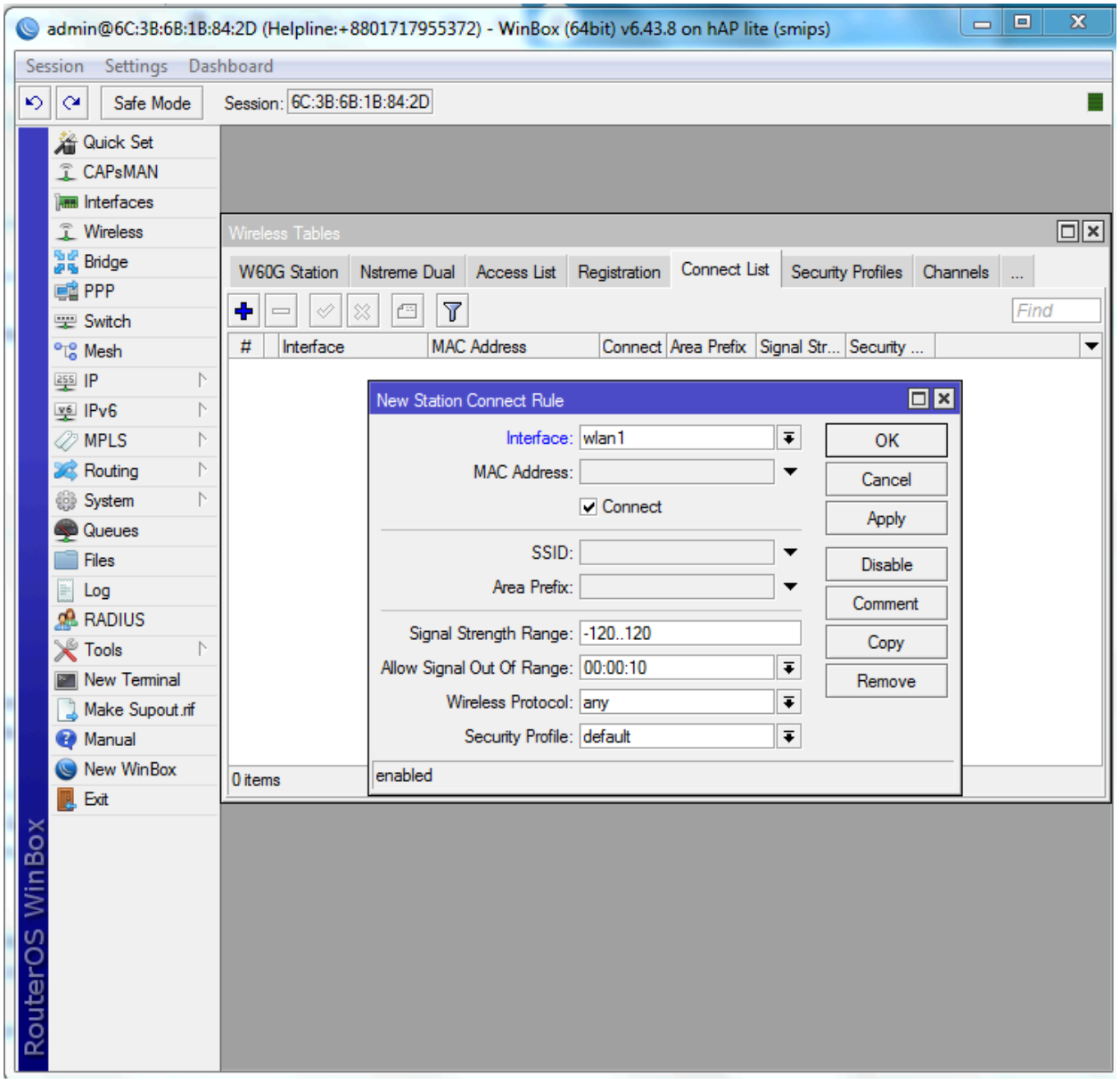
Operation:

Access list rules are checked sequentially.
Disabled rules are always ignored.
Only the first matching rule is applied.
If there are no matching rules for the remote connection, then the default values from the wireless interface configuration are used.
If remote device is matched by rule that has authentication=no value, the connection from that remote device is rejected.



Connect-List

Sub-menu: /interface wireless connect-list



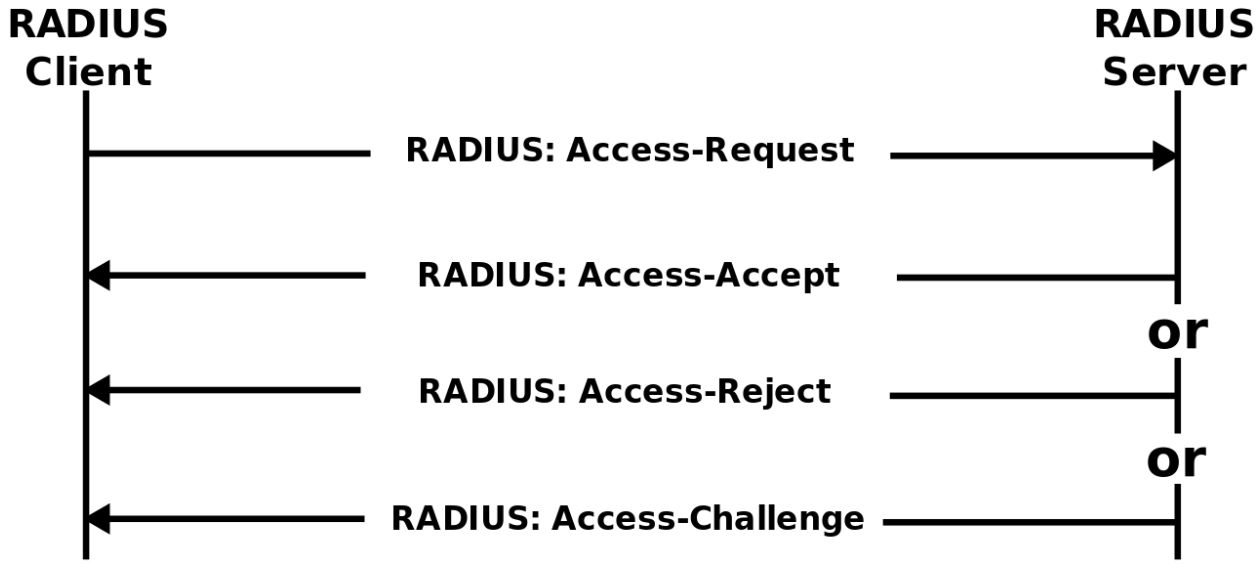
connect-list is used to assign priority and security settings to connections with remote access points, and to restrict allowed connections. connect-list is an ordered list of rules. Each rule in connect-list is attached to specific wireless interface, specified in the interface property of that rule (this is unlike access-list, where rules can apply to all interfaces). Rule can match MAC address of remote access point, it's signal strength and many other parameters.

Operation:

connect-list rules are always checked sequentially, starting from the first.
disabled rules are always ignored.

Only the first matching rule is applied.
If SSID or exact wireless protocol is provided in the wireless interface configuration Connect List SSIDs or wireless protocols not covered by wireless interface configuration are ignored.
If connect-list does not have any rule that matches remote access point, then the default values from the wireless interface configuration are used.
If access point is matched by rule that has connect=no value, connection with this access point will not be attempted.
If access point is matched by rule that has connect=yes value, connection with this access point will be attempted.
In station mode, if several remote access points are matched by connect list rules with connect=yes value, connection will be attempted with access point that is matched by rule higher in the connect-list.
If no remote access points are matched by connect-list rules with connect=yes value, then value of default-authentication interface property determines whether station will attempt to connect to any access point. If default-authentication=yes, station will choose access point with best signal and compatible security.
In access point mode, connect-list is checked before establishing WDS link with remote device. If access point is not matched by any rule in the connect list, then the value of default-authentication determines whether WDS link will be established.

RADIUS
Remote Authentication Dial-In User Service (RADIUS) is a networking protocol, operating on port 1812, that provides centralized Authentication, Authorization, and Accounting (AAA or Triple A) management for users who connect and use a network service. RADIUS was developed by Livingston Enterprises, Inc. in 1991 as an access server authentication and accounting protocol and later brought into the Internet Engineering Task Force (IETF) standards.

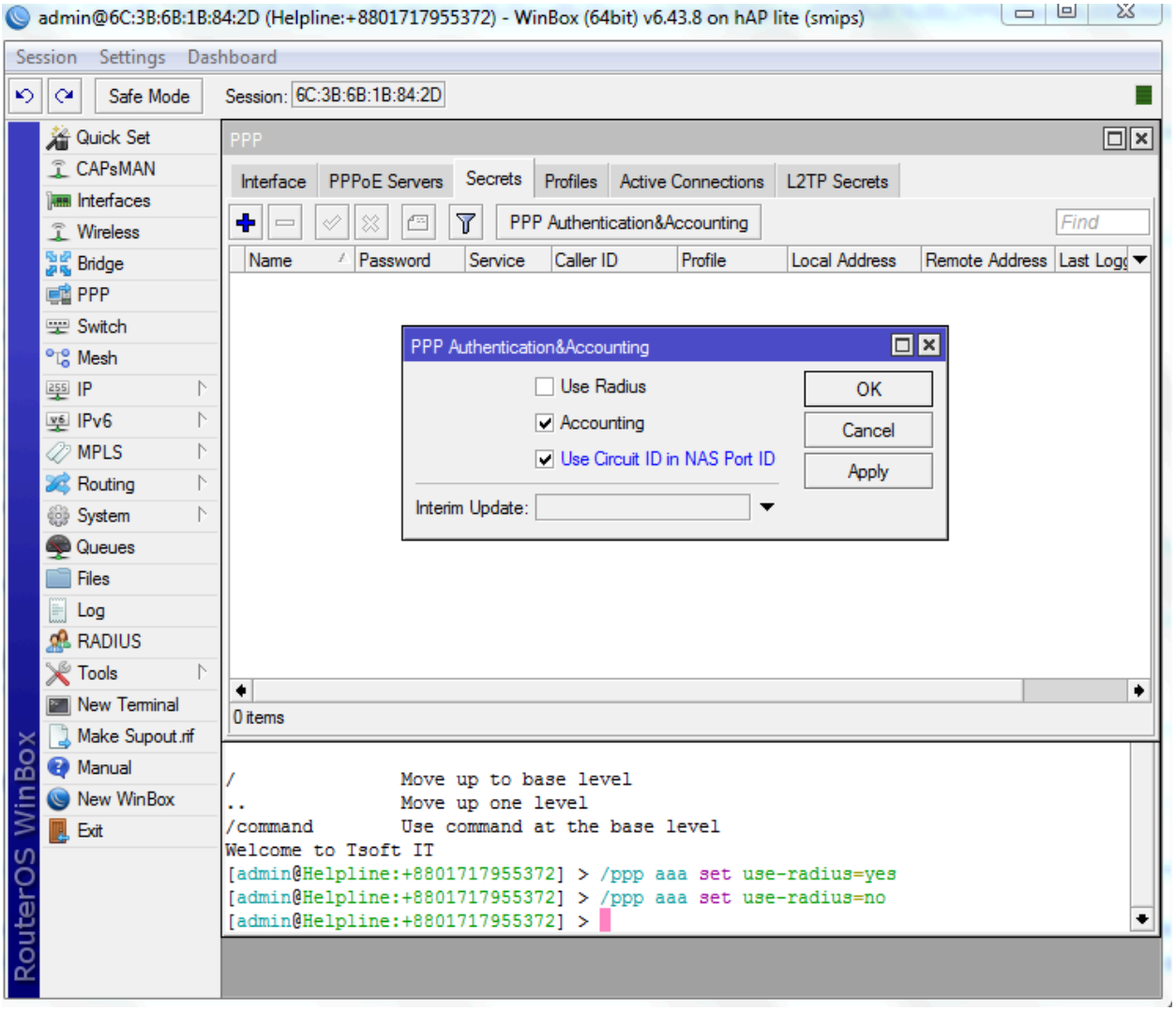


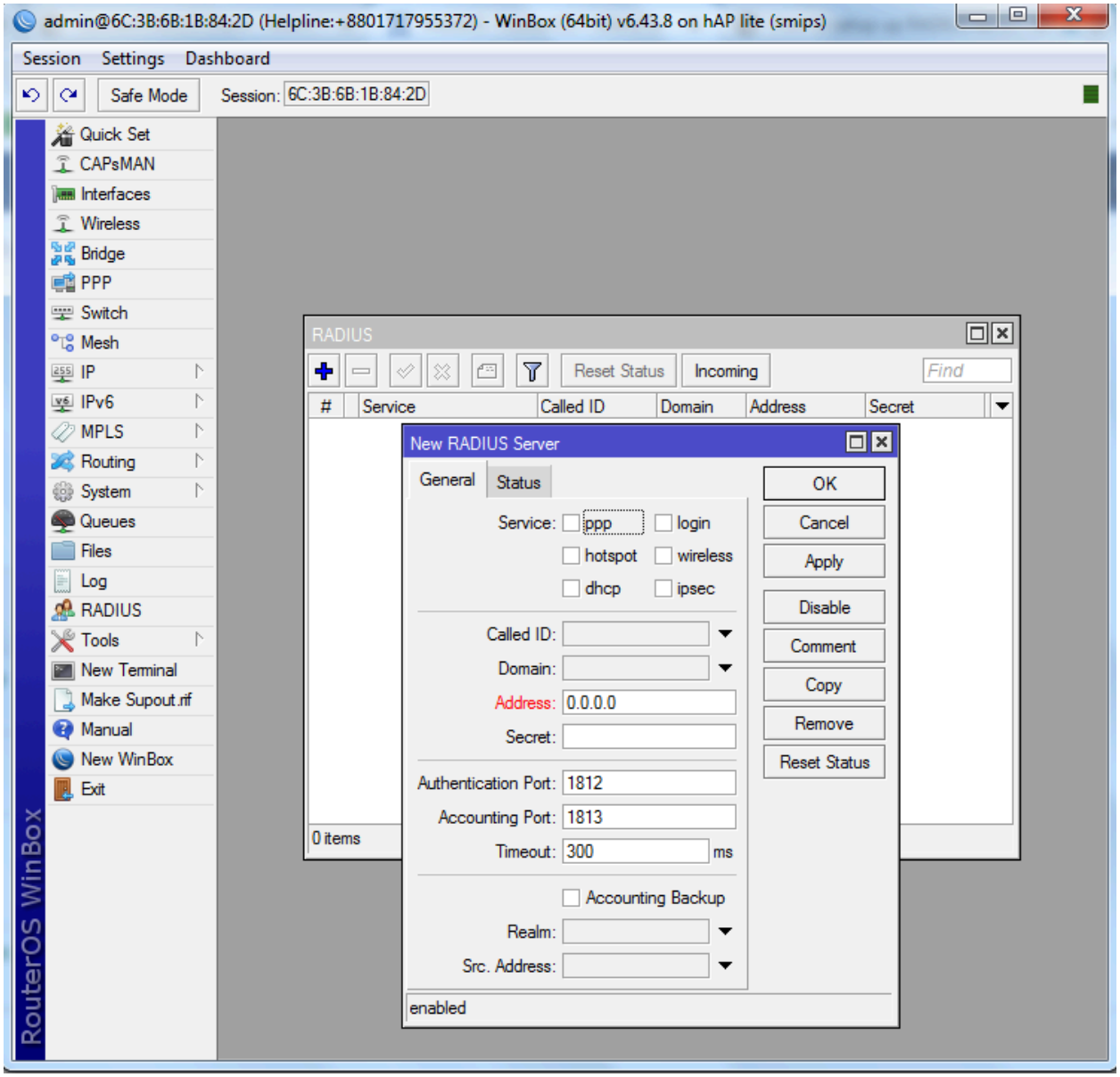
Access Reject

The user is unconditionally denied access to all requested network resources. Reasons may include failure to provide proof of identification or an unknown or inactive user account.
Access Challenge
Requests additional information from the user such as a secondary password, PIN, token, or card. Access Challenge is also used in more complex authentication dialogs where a secure tunnel is established between the user machine and the Radius Server in a way that the access credentials are hidden from the NAS.
Access Accept
The user is granted access. Once the user is authenticated, the RADIUS server will often check that the user is authorized to use the network service requested. A given user may be allowed to use a company's wireless network, but not its VPN service, for example. Again, this information may be stored locally on the RADIUS server, or may be looked up in an external source such as LDAP or Active Directory.

RADIUS authentication and accounting gives the ISP or network administrator ability to manage PPP user access and accounting from one server throughout a large network. The MikroTik RouterOS has a RADIUS client which can authenticate for HotSpot, PPP, PPPoE, PPTP, L2TP and ISDN connections.

Radius server enable


admin@6C:3B:6B:1B:84:2D (Helpline:+8801717955372) - WinBox (64bit) v6.43.8 on hAP lite (smips)
Session Settings Dashboard
Safe Mode
Session: 6C:3B:6B:1B:84:2D
Quick Set
CAPsMAN
Interfaces
Wireless
Bridge
PPP
Switch
Mesh
IP
IPv6
MPLS
Routing
System
Queues
Files
Log
RADIUS
Tools
New Terminal
Make Supout.rif
Manual
New WinBox
Exit
RouterOS WinBox
PPP
Interface
PPPoE Servers
Secrets
Profiles
Active Connections
L2TP Secrets
PPP Authentication&Accounting
Find
Name
Password
Service
Caller ID
Profile
Local Address
Remote Address
Last Log
PPP Authentication&Accounting
Use Radius
Accounting
Use Circuit ID in NAS Port ID
Interim Update:
OK
Cancel
Apply
0 items
/ Move up to base level
.. Move up one level
/command Use command at the base level
Welcome to Tsoft IT
[admin@Helpline:+8801717955372] > /ppp aaa set use-radius=yes
[admin@Helpline:+8801717955372] > /ppp aaa set use-radius=no
[admin@Helpline:+8801717955372] >



WLAN Authentication Methods

It is important to understand that there is a distinction between being authenticated onto a wireless network and then having the traffic passed be encrypted. It is possible to be authenticated onto a network and pass open unencrypted traffic; this section looks at the commonly used methods of authentication.

There are three main methods of authentication that are used on today's wireless LANs:

open authentication
shared authentication
EAP (Extensible Authentication Protocol) authentication

The open authentication method is the simplest of the methods used and only requires that the end device be aware of the Service-Set Identifier (SSID) used on the network, as long as the SSID is known then the device will be allowed onto the network. The problem with this method is that the SSID is typically broadcast and if it is not, it can be easy to figure out with passive capturing techniques.

The shared authentication method is commonly used on individual and small business wireless LAN implementations; this method uses a shared key (Pre-Shared Key – PSK) that is given to both sides of the connection; if they match then the device is allowed onto the network.

The third method uses the Extensible Authentication Protocol (EAP) and is the most common method used by enterprises. The EAP method utilizes an authentication server that is queried for authentication using a variety of credential options.

WLAN Encryption Methods
Along with the method used for authentication, the choice of encryption method is a very important part of deploying a wireless LAN. Many of the encryption methods that were implemented in earlier wireless LAN standards have been proven insecure and have been depreciated by more modern methods. As time goes on, this is sure to happen with all encryption techniques as they are used more commonly (thus becoming a target for exploitation) and as processing power continues to increase.

Here are the WLAN encryption methods we'll review today:

Wired Equivalent Privacy (WEP)
Wi-Fi Protected Access (WPA)
Wi-Fi Protected Access 2 (WPA2)
The first widely used standard for wireless LANs was 802.11 (prime); this included the Wired Equivalent Privacy (WEP) algorithm which was used for security. WEP utilizes RC4 for encryption and has been depreciated because of vulnerabilities that can be used to find the security keys.

In response to the vulnerabilities found in WEP, Wi-Fi Protected Access (WPA) was defined. WPA utilizes the Temporal Key Integrity Protocol (TKIP) which utilizes dynamic keys that were not supported with WEP and RC4 for encryption. The TKIP method used with WPA was utilized until vulnerabilities were found in TKIP. These vulnerabilities center on the fact that TKIP uses some of the same mechanisms that WEP does which allow similar attacks.

In response to the vulnerabilities in WPA/TKIP, the IEEE 802.11i standard was defined and implemented; the IEEE 802.11i standard is also referred to as WPA2. WPA2 replaced TKIP with Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP) which is based on Advanced Encryption Standard (AES); it is common for the WPA2 encryption

method to be referred to as AES. As of this writing, there are no easy methods that have been found to break AES.

EAP
Extensible Authentication Protocol (EAP) is an authentication framework frequently used in network and internet connections.
EAP is in wide use. For example, in IEEE 802.11 (WiFi) the WPA and WPA2 standards have adopted IEEE 802.1X (with various EAP types) as the canonical authentication mechanism.

Management Frame Protection
Management Frame Protection (MFP) is a wireless feature that increases the security of the management frames. Its wireless standard is IEEE 802.11w-2009 or Protected Management Frames (PMF) which aims to provide data confidentiality of the management frames and protect wireless connectivity.

Wireless WDS protocol
A wireless distribution system (WDS) is a system enabling the wireless interconnection of access points in an IEEE 802.11 network. It allows a wireless network to be expanded using multiple access points without the traditional requirement for a wired backbone to link them.

Sub-menu: /interface wireless wds

Dynamic/Static WDS
With static WDS modes you willl add a WDS interface and enter the MAC address of the remote AP. With dynamic WDS modes you don't have to, but probably will, enter the MAC address of the remote AP under the connect list as per the beginning of this thread.
static is useful in the special case of a ring configuration between 4 points(A-B-C-D) and you need only to connect in a sequential scheme, avoiding casuals wds connection between A - C and B - D.

• Use dynamic/dynamic-mesh WDS mode only in
the beginning when creating the WDS MESH
setup
• Having dynamic WDS interfaces with low signal
levels and bad traffic throughput could cause
WDS link disconnect and that causes the bridge
tree rebuild
• Use connect-list to disallow making WDS links
with bad signal or convert dynamic WDS
interfaces to static and switch to static/staticmesh WDS mode

RSTP Bridge
Rapid Spanning Tree Protocol
STP and Rapid STP are used very widely across many networks, but almost all networks have switched over using only RSTP since of its benefits. STP is a very old protocol and has a convergence time (the time needed to fully learn network topology changes and to continue properly forwarding traffic) even up to 50 seconds, which was acceptable for the 1980s when it was invented. RSTP has a lot of smaller convergence time, a few seconds or even a few milliseconds), which is acceptable for nowadays network requirements. It is recommended to use RSTP instead of STP since it is a lot faster and is also backward compatible with STP. One of the reasons why RSTP is faster is because of reduced possible port states, below is a list of possible STP port states:

Forwarding - port participates in traffic forwarding and is learning MAC addresses, is receiving BPDUs.
Listening - port does not participate in traffic forwarding and is not learning MAC addresses, is receiving BPDUs.
Learning - port does not participate in traffic forwarding but is learning MAC addresses.
Blocking - port is blocked since it is causing loops but is receiving BPDUs.
Disabled - port is disabled or inactive.
In RSTP the disabled, listening and blocking port states are replaced with just one state called the Discarding state:

Forwarding - port participates in traffic forwarding and is learning MAC addresses, is receiving BPDUs (forwarding=yes).
Learning - port does not participate in traffic forwarding but is learning MAC addresses (learning=yes).
Discarding - port does not participate in traffic forwarding and is not learning MAC addresses, is receiving BPDUs (forwarding=no).

```
/interface bridge
add name=bridge protocol-mode=rstp vlan-filtering=no
/interface bridge port
add bridge=bridge interface=ether1
add bridge=bridge interface=ether2
add bridge=bridge interface=ether3
add bridge=bridge interface=ether4
```

Wireless Mesh
HWMP+ is a MikroTik specific layer-2 routing protocol for wireless mesh networks. It is based on Hybrid Wireless Mesh Protocol (HWMP) from IEEE 802.11s draft standard. It can be used instead of (Rapid) Spanning Tree protocols in mesh setups to ensure loop-free optimal routing.

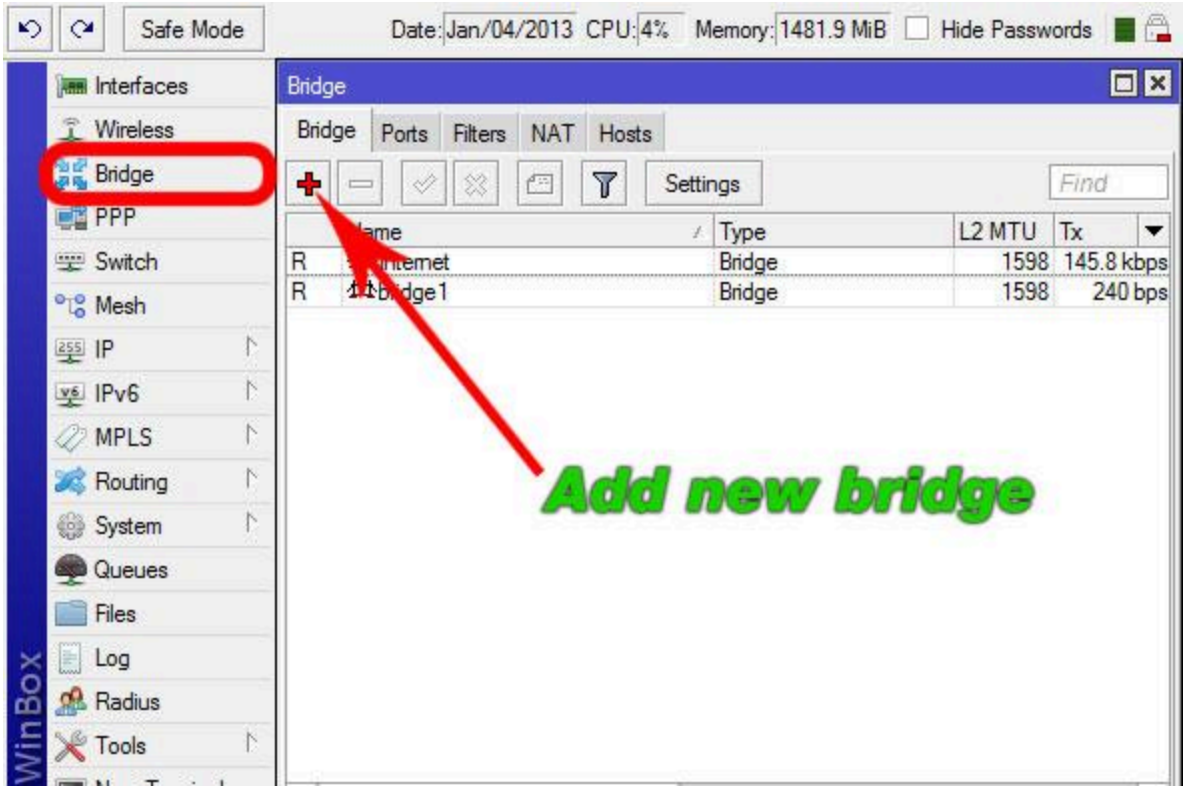
The HWMP+ protocol however is not compatible with HWMP from IEEE 802.11s draft standard.

Note that the distribution system you use for your network need not to be Wireless Distribution System (WDS). HWMP+ mesh routing supports not only WDS interfaces, but also Ethernet interfaces inside the mesh. So you can use simple Ethernet based distribution system, or you can combine both WDS and Ethernet links!
Sub-menu: /interface mesh port

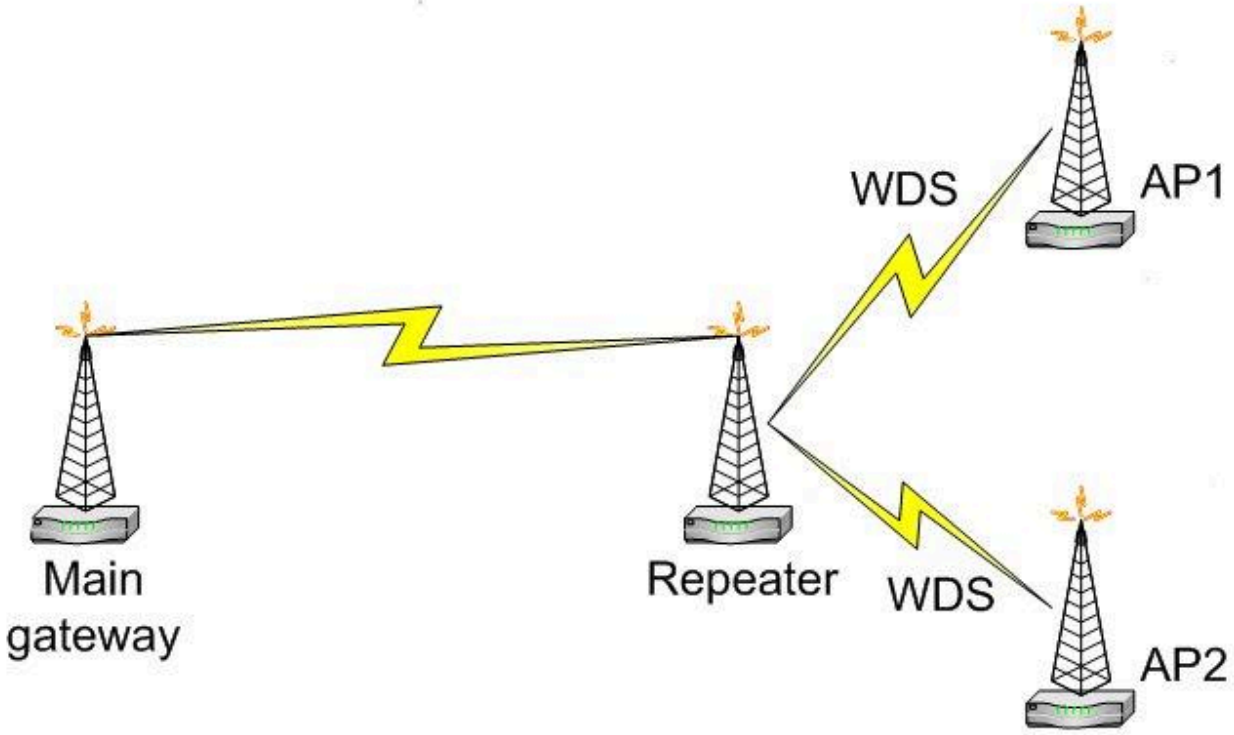
Wireless Bridging

Bridges are using to connect network segments together in layer 2. As I said almost switches acting as bridges today and don't need to configure anything on them. But a Mikrotik router (unlike a cisco switch) won't work as bridge automatically and need to be configure

How to configure Bridge on a Mikrotik router?



WDS bridging
This example shows how to configure a wireless repeater. Wireless repeater extends the range of an existing WLAN instead of adding more access points. Consider the network layout:



We will use two wireless interfaces (two antennas) on the repeater router. WDS links will be established between 'Main gateway' and 'Repeater', 'Repeater' and 'AP1', 'AP2' (end-users are connected to the AP1 and AP2).

Note, client wireless interfaces (station, ad-hoc, infrastructure) do not support bridging because of the limitations of 802.11.

Quick Start
Main Gateway configuration export:

```
/ ip address add address=192.168.0.1/24 interface=wlan1
/ interface wireless set wlan1 disabled=no mode=ap-bridge band=5ghz frequency=5180
ssid=Main_gw wds-mode=static
/ interface wireless wds add disabled=no wds-address=XX:XX:XX:XX:XX:X2
master-interface=wlan1
/ interface bridge add
/ interface bridge port add interface=wlan1 bridge=bridge1
```

/ interface bridge port add interface=wds1 bridge=bridge1
Repeater configuration export:

/ interface wireless set wlan1 disabled=no mode=ap-bridge band=5ghz frequency=5180 ssid=Main_gw wds-mode=static
/ interface wireless wds add disabled=no wds-address=XX:XX:XX:XX:XX:X1 master-interface=wlan1
/ interface bridge add
/ interface bridge port add interface=wlan1 bridge=bridge1
/ interface bridge port add interface=wds1 bridge=bridge1

/ interface wireless set wlan2 disabled=no mode=ap-bridge band=5ghz frequency=5805 ssid=To_clients wds-mode=static
/ interface wireless wds add disabled=no wds-address=XX:XX:XX:XX:XX:X4 master-interface=wlan2
/ interface bridge port add interface=wlan2 bridge=bridge1
/ interface bridge port add interface=wds2 bridge=bridge1
/ ip address add address=192.168.0.2/24 interface=bridge1
AP1, AP2 configuration export:

/ interface wireless set wlan1 disabled=no mode=ap-bridge band=5ghz frequency=5805 ssid=To_clients wds-mode=static
/ interface wireless wds add disabled=no wds-address=XX:XX:XX:XX:XX:X3 master-interface=wlan1
/ interface bridge add
/ interface bridge port add interface=wlan1 bridge=bridge1
/ interface bridge port add interface=wds1 bridge=bridge1
/ ip adress add address=192.168.0.3/24 interface=wlan1
Explanation
Main Gateway
/ ip address add address=192.168.0.1/24 interface=wlan1
The wireless interface has the name 'wlan1' and IP address of 192.168.0.1/24. It will be the default gateway for other nodes in the network.

/ interface wireless set wlan1 disabled=no mode=ap-bridge band=5ghz frequency=5180 ssid=Main_gw wds-mode=static
Rule enables wireless interface and sets up appropriate configuration: band, frequency (WDS peers all need to be on the same frequency), SSID, wds-mode (static - WDS interfaces created manually in static mode).

/ interface wireless wds add disabled=no wds-address=XX:XX:XX:XX:XX:X2 master-interface=wlan1

The rule creates WDS interface on the particular 'master-interface', where 'wds-address' is the remote WDS peer MAC-address.

/ interface bridge add
/ interface bridge port add interface=wlan1 bridge=bridge1
/ interface bridge port add interface=wds1 bridge=bridge1
First we create bridge interface on the router, then 'wlan1' and 'wds1' interface are added to the bridge.

Repeater
Let us consider wireless repeater configuration. First we need to establish communication to 'Main Gateway' router.

/ interface wireless set wlan1 disabled=no mode=ap-bridge band=5ghz frequency=5180 ssid=Main_gw wds-mode=static
/ interface wireless wds add disabled=no wds-address=XX:XX:XX:XX:XX:X1 master-interface=wlan1
/ interface bridge add
/ interface bridge port add interface=wlan1 bridge=bridge1
/ interface bridge port add interface=wds1 bridge=bridge1
The rules are quite similar to 'Main Gateway' configuration rules. We need to enable wireless interface, than set appropriate wireless and WDS configuration (don't forget about the same frequency channel) and add interface to created bridge.

/ interface wireless set wlan2 disabled=no mode=ap-bridge band=5ghz frequency=5805 ssid=To_clients wds-mode=static
/ interface wireless wds add disabled=no wds-address=XX:XX:XX:XX:XX:X4 master-interface=wlan2
/ interface bridge port add interface=wlan2 bridge=bridge1
/ interface bridge port add interface=wds2 bridge=bridge1
/ ip address add address=192.168.0.2/24 interface=bridge1
Next, we should configure second wireless interface. Rules are very similar to the previous one. IP address configuration is required for troubleshoot options. Repetear will work fine without IP address configuration.

AP1, AP2 configuration
/ interface wireless set wlan1 disabled=no mode=ap-bridge band=5ghz frequency=5805 ssid=To_clients wds-mode=static
/ interface wireless wds add disabled=no wds-address=XX:XX:XX:XX:XX:X3 master-interface=wlan1
/ interface bridge add
/ interface bridge port add interface=wlan1 bridge=bridge1
/ interface bridge port add interface=wds1 bridge=bridge1

/ ip adress add address=192.168.0.3/24 interface=wlan1
Finally, we need to configure AP1, AP2 client routers, which will provide connection to the Internet for the end-user. Add client interface to bridge (Ethernet, wireless). AP1 and AP2 clients will use 192.168.0.0/24 subnet, AP1 and AP2 clients will communicate as they are connected to the one switch.

Mode station-wds
This mode works only with RouterOS APs. As a result of negotiating connection, separate WDS interface is created on AP for given station. This interface can be thought of point-to-point connection between AP and given station - whatever is sent out WDS interface is delivered to station (and only to particular station) and whatever station sends to AP is received from WDS interface (and not subject to forwarding between AP clients), preserving L2 addresses.

This mode is supported for all wireless protocols except when 802.11 protocol is used in connection to non-RouterOS device. Mode uses 4 address frame format when used with 802.11 protocol, for other protocols (such as nstreme or nv2), protocol internal means are used.

This mode is safe to use for L2 bridging and gives most administrative control on AP by means of separate WDS interface, for example use of bridge firewall, RSTP for loop detection and avoidance, etc.

With station-wds mode, it is not possible to connect to CAPsMAN controlled CAP.

Mode station-pseudobridge
From the wireless connection point of view, this mode is the same as standard station mode. It has limited support for L2 bridging by means of some services implemented in station:

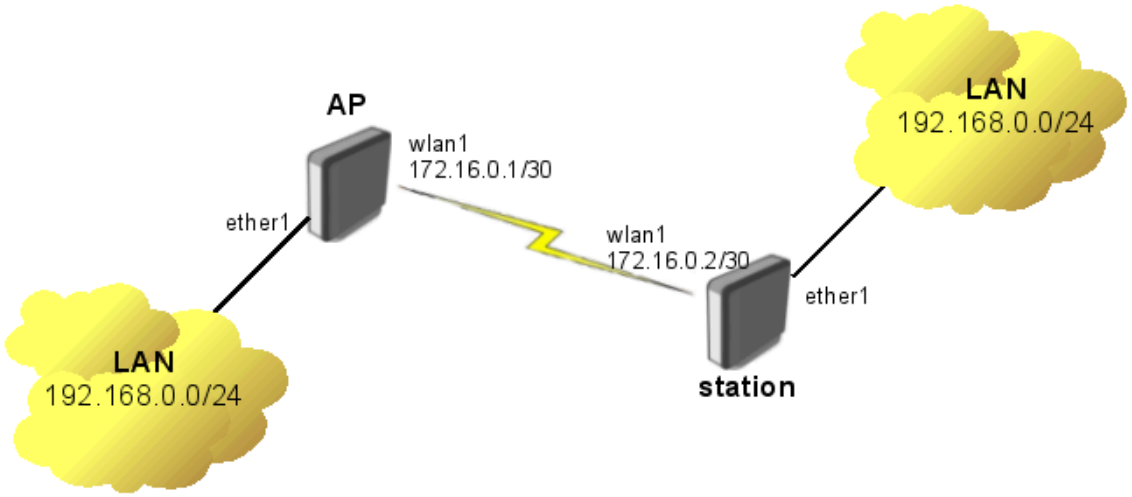
MAC address translation for IPv4 packets - station maintains IPv4-to-MAC mapping table and replaces source MAC address with its own address when sending frame to AP (in order to be able to use 3 address frame format), and replaces destination MAC address with address from mapping table for frames received from AP. IPv4-to-MAC mappings are built also for VLAN encapsulated frames.
single MAC address translation for the rest of protocols - station learns source MAC address from first forwarded non-IPv4 frame and uses it as default for reverse translation - this MAC address is used to replace destination MAC address for frames received from AP if IPv4-to-MAC mapping can not be performed (e.g. - non-IPv4 frame or missing mapping).
This mode is limited to complete L2 bridging of data to single device connected to station (by means of single MAC address translation) and some support for IPv4 frame bridging - bridging of non-IP protocols to more than one device will not work. Also MAC address translation limits access to station device from AP side to IPv4 based access - the rest of protocols will be translated by single MAC address translation and will not be received by station itself.

This mode is available for all protocols except nv2 and should be avoided when possible. The usage of this mode can only be justified if AP does not support better mode for L2 bridging (e.g. when non-RouterOS AP is used) or if only one end-user device must be connected to network by means of station device.

MPLS/VPLS tunnel
VPLS- Virtual Private LAN Service
Multiprotocol Label Switching (MPLS)

The MPLS/VPLS approach has some advantages:
VPLS tunnel is about 60% faster and less overhead than EoIP tunnel
802.11n speed is limited over WDS bridges, this method doesn't have such limitations

Let us assume the following network setup:



Note: For this setup to work in RouterOS 3.x, routing and mpls-test package must be installed. These features are included by default in 4.x.

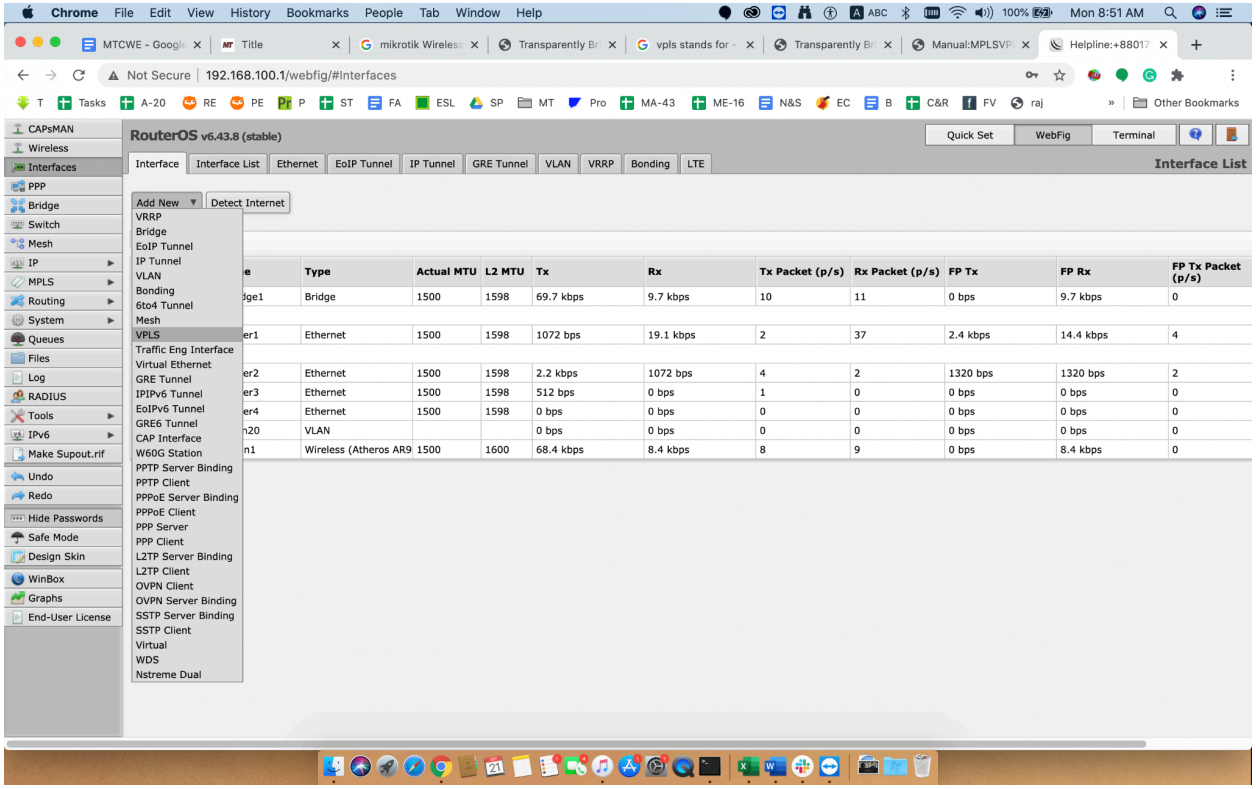
AP

```
# --configure wireless access point--
/interface wireless
set wlan1 disabled=no ssid=MPLS frequency=5180 band=5ghz mode=bridge

# --configure IP--
/ip address
add address=172.16.0.1/30 interface=wlan1
```

--enable LDP--
/mpls ldp
set enabled=yes lsr-id=172.16.0.1 transport-address=172.16.0.1
/mpls ldp interface
add interface=wlan1

--configure VPLS tunnel--



/interface vpls
add name=vpls1 remote-peer=172.16.0.2 vpls-id=1:1 disabled=no

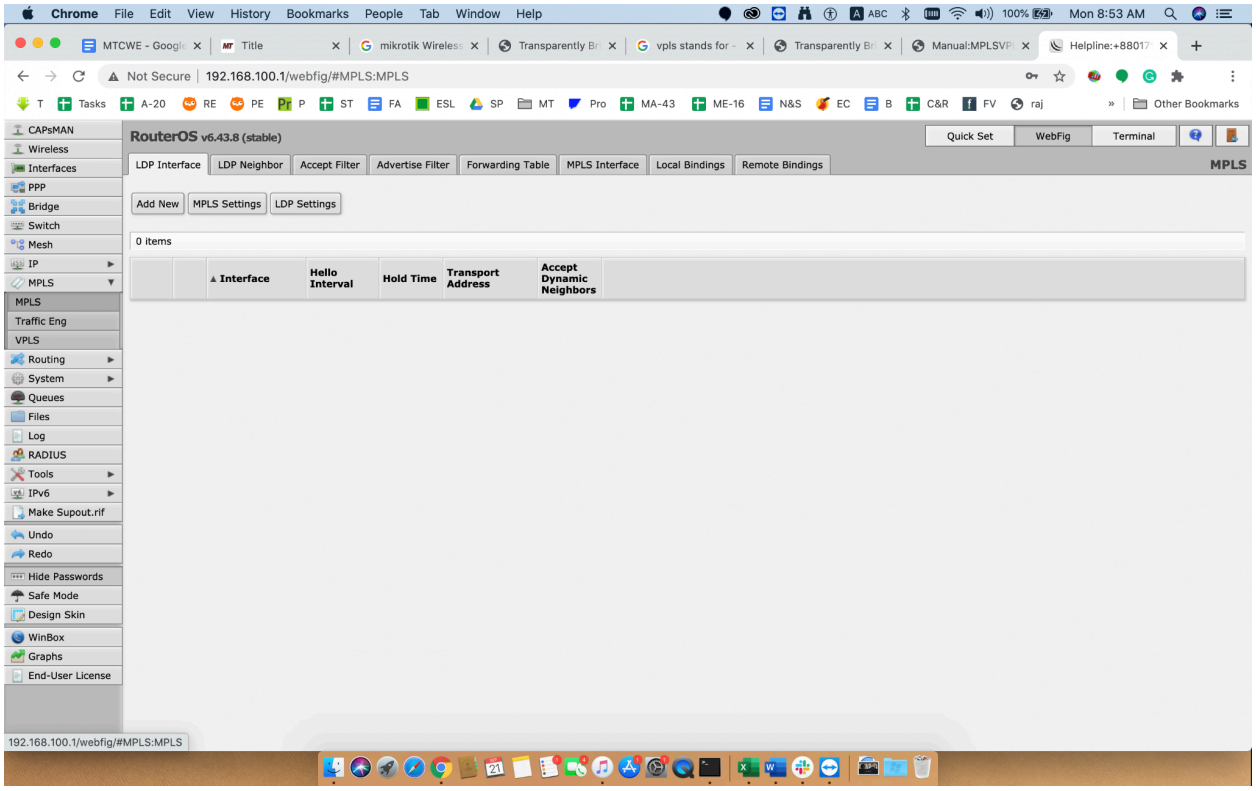
--add bridge and bridge ports --
/interface bridge add name=bridge1
/interface bridge port
add bridge=bridge1 interface=ether1
add bridge=bridge1 interface=vpls1
station

--configure wireless access point--
/interface wireless
set wlan1 disabled=no ssid=MPLS band=5ghz mode=station

--configure IP--

/ip address
add address=172.16.0.2/30 interface=wlan1

--enable LDP--



/mpls ldp
set enabled=yes lsr-id=172.16.0.2 transport-address=172.16.0.2
/mpls ldp interface
add interface=wlan1

--configure VPLS tunnel--
/interface vpls
add name=vpls1 remote-peer=172.16.0.1 vpls-id=1:1 disabled=no

--add bridge and bridge ports --
/interface bridge add name=bridge1
/interface bridge port
add bridge=bridge1 interface=ether1
add bridge=bridge1 interface=vpls1
Confirm that LDP is running

[admin@MikroTik] /mpls ldp neighbor> print
Flags: X - disabled, D - dynamic, O - operational, T - sending-targeted-hello, V - vpls
 # TRANSPORT LOCAL-TRANSPORT PEER SEND-TARGETED ADDRESSES

```
 0 DOTV 172.16.0.2   172.16.0.1      172.16.0.2:0   no           172.16.0.2

[admin@MikroTik] /mpls forwarding-table print
Flags: L - ldp, V - vpls, T - traffic-eng
 #   IN-LABEL       OUT-LABELS      DESTINATION    INTERFACE    NEXTHOP
 0   expl-null
 1 V 18                             vpls1
```

Confirm that VPLS tunnel is established:

```
[admin@MikroTik] /interface vpls> monitor vpls1 once
       remote-label: 17
        local-label: 18
      remote-status:
  transport-nexthop: 172.16.0.2
     imposed-labels: 17
```

Connect more stations to the same Bridge

It is possible to connect more than one Station to the same Bridge, by simply creating more VPLS tunnels (one for Station, also don't forget to increase subnet size from /30 to /24 for example, depending how much clients you need):

AP

```
# --change wireless access point mode to AP Bridge--
/interface wireless
set wlan1 mode=ap-bridge

# --configure second VPLS tunnel--
/interface vpls
add name=vpls2 remote-peer=172.16.0.3 vpls-id=2:2 disabled=no

# --add vpls2 to bridge --

/interface bridge port
add bridge=bridge1 interface=vpls2
```

2nd Station

```
# --configure wireless access point--
/interface wireless
set wlan1 disabled=no ssid=MPLS band=5ghz mode=station
```

```
# --configure IP--
/ip address
add address=172.16.0.3/30 interface=wlan1

# --enable LDP--
/mpls ldp
set enabled=yes lsr-id=172.16.0.3 transport-address=172.16.0.3
/mpls ldp interface
add interface=wlan1

# --configure VPLS tunnel--
/interface vpls
add name=vpls1 remote-peer=172.16.0.1 vpls-id=2:2 disabled=no

# --add bridge and bridge ports --
/interface bridge add name=bridge1
/interface bridge port
add bridge=bridge1 interface=ether1
add bridge=bridge1 interface=vpls1
```

MikroTik Wireless Nstreme Protocol

Nstreme
Sub-menu: /interface wireless nstreme

This menu allows to switch a wireless card to the nstreme mode. In this case the card will work only with nstreme clients.

Nstreme Dual
Sub-menu: /interface wireless nstreme-dual

Two radios in nstreme-dual-slave mode can be grouped together to make nstreme2 Point-to-Point connection. To put wireless interfaces into a nstreme2 group, you should set their mode to nstreme-dual-slave. Many parameters from /interface wireless menu are ignored, using the nstreme2, except:

frequency-mode
country

antenna-gain
tx-power
tx-power-mode
Antenna-mode

Nv2 protocol is proprietary wireless protocol developed by MikroTik for use with Atheros 802.11 wireless chips. Nv2 is based on TDMA (Time Division Multiple Access) media access technology instead of CSMA (Carrier Sense Multiple Access) media access technology used in regular 802.11 devices.

TDMA media access technology solves hidden node problem and improves media usage, thus improving throughput and latency, especially in PtMP networks.

Nv2 is supported for Atheros 802.11n chips and legacy 802.11a/b/g chips starting from AR5212, but not supported on older AR5211 and AR5210 chips. This means that both - 11n and legacy devices can participate in the same network and it is not required to upgrade hardware to implement Nv2 in network.

Media access in Nv2 network is controlled by Nv2 Access Point. Nv2 AP divides time in fixed size "periods" which are dynamically divided in downlink (data sent from AP to clients) and uplink (data sent from clients to AP) portions, based on queue state on AP and clients. Uplink time is further divided between connected clients based on their requirements for bandwidth. At the begginning of each period AP broadcasts schedule that tells clients when they should transmit and the amount of time they can use.

In order to allow new clients to connect, Nv2 AP periodically assigns uplink time for "unspecified" client - this time interval is then used by fresh client to initiate registration to AP. Then AP estimates propagation delay between AP and client and starts periodically scheduling uplink time for this client in order to complete registration and receive data from client.

Nv2 implements dynamic rate selection on per-client basis and ARQ for data transmissions. This enables reliable communications across Nv2 links.

For QoS Nv2 implements variable number of priority queues with builtin default QoS scheduler that can be accompanied with fine grained QoS policy based on firewall rules or priority information propagated across network using VLAN priority or MPLS EXP bits.

Nv2 protocol limit is 511 clients per interface.

Configuration example
Master AP:

```
/interface wireless set wlan1 mode=ap-bridge ssid=Sector1 frequency=5220
nv2-mode=sync-master nv2-preshared-key=clients1 nv2-sync-secret=Tower1
```

Slave AP:

```
/interface wireless set wlan1 mode=ap-bridge ssid=Sector2 frequency=5220
nv2-mode=sync-slave nv2-preshared-key=clients2 nv2-sync-secret=Tower1
```

Monitor interface on the Slave AP:

```
[admin@SlaveAP] /interface wireless> monitor wlan1
                  status: running-ap
                 channel: 5220/20/an
       wireless-protocol: nv2
             noise-floor: -110dBm
      registered-clients: 1
    authenticated-clients: 1
          nv2-sync-state: synced
         nv2-sync-master: 4C:5E:0C:57:84:38
        nv2-sync-distance: 1
     nv2-sync-period-size: 2
  nv2-sync-downlink-ratio: 50
```

Debug logs on the Master AP:

```
09:22:08 wireless,debug wlan1: 4C:5E:0C:57:85:BE attempts to sync
```

Debug logs on the Slave AP:

```
09:22:08 wireless,debug wlan1: attempting to sync to 4C:5E:0C:57:84:38
09:22:09 wireless,debug wlan1: synced to 4C:5E:0C:57:84:38
```